

Practice Guideline



AASLD AST Practice Guideline on adult liver transplantation: Diagnosis and management of graft-related complications

Abstract

Background and Aim: Advances in immunosuppression and surgical techniques in liver transplantation (LT) have significantly improved patient outcomes, but donor utilization and access to LT remain a challenge. The past decade has seen significant strides in donor pool expansion with acceptable clinical outcomes, while improved patient selection and advances in the management of chronic liver disease and post-LT complications have promoted better allograft health. This document aims to provide an evidence-based guideline on the management of graft complications, immunosuppression, graft rejection and recurrent disease in adult LT recipients.

Methods: A multidisciplinary writing group of experts (transplant hepatologists, surgeons, pathologist, and psychiatrist) was tasked to formulate clinical questions (in PICO format) that arise during routine management of adult LT recipients. The writing group reviewed the literature, generated guideline recommendations and rated the level of evidence for each recommendation based on the Oxford Center for Evidence-Based Medicine. The group categorized the strength of recommendations based on the level of evidence, risk-benefit ratio, and patient preferences.

Conclusion: While robust clinical trial data provide strong guidance on some aspects of graft management in LT such as machine perfusion and immunosuppression efficacy and safety, a significant component of graft management is derived from retrospective cohort data, extrapolation of data from other solid organ

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Abbreviations: A2ALL, Adult-to-Adult Living Donor Transplant Cohort Study; AAH, acute alcohol-associated hepatitis; AASLD, American Association for the Study of Liver Diseases; ACR, acute cellular rejection; AIH, autoimmune hepatitis; ALD, alcohol-associated liver disease; AMR, antibody-mediated rejection; APRI, AST-to-platelet ratio index; AST, American Society of Transplantation; AUD, alcohol use disorder; AZA, azathioprine; BAS, basiliximab; BPAR, biopsy-proven acute rejection; C4d, complement 4d; CD, Clavien-Dindo; CDR, chronic ductopenic rejection; CNi, calcineurin inhibitor; CSA, cyclosporine; CTA, computed tomographic angiography; CTV, computed tomographic venography; CYP3A, cytochrome P-450 3A; CYP3A4, cytochrome P450 3A4; DAA, direct-acting antiviral; DAC, daclizumab; DBD, donation after brain death; DCD, donation after circulatory death; DDLT, deceased donor liver transplant; DM, diabetes mellitus; DSA, donor-specific antibodies; dTAC, delayed tacrolimus; EAD, early allograft dysfunction; EASE, Early Allograft Failure Simplified Estimation; ELTR, European Liver Transplant Registry; ERCP, endoscopic retrograde cholangiopancreatography; EVL, everolimus; FIB-4, fibrosis score-4; GFR, glomerular filtration rate; HAI, hepatic activity index; HAT, hepatic artery thrombosis; HBIG, hepatitis B immunoglobulin; HLA, human leukocyte antigen; HMP, hypothermic machine perfusion; HRAR, High-Risk Alcohol Relapse Scale; HTN, hypertension; IFN, interferon; IgG4, immunoglobulin G4; IL2-RA, interleukin-2 receptor antagonists; IMV, inferior mesenteric vein; IR, interventional radiology; LDLT, living donor liver transplant; L-Graft, Liver Graft Assessment Following Transplantation; LT, liver transplantation; MASH, metabolic dysfunction-associated steatohepatitis; MASLD, metabolic dysfunction-associated steatotic liver disease; MDRD, modification of diet in renal disease; MEAF, Model for Early Allograft Function; MFI, mean fluorescence intensity; miRNA, microRNA; MMF, mycophenolate mofetil; MRA, magnetic resonance angiography; MRCP, magnetic resonance cholangiopancreatography; MRV, magnetic resonance venography; mTOR, mammalian target of rapamycin; NA, nucleos(t)ide analogue; NABS, non-anastomotic biliary strictures; NAC, N-acetylcysteine; NAT, nucleic acid test; NMP, normothermic machine perfusion; NRP, normothermic regional perfusion; OPTN, Organ Procurement and Transplantation Network; OS, overall survival; PBC, primary biliary cholangitis; PCR, plasma cell-rich rejection; PEth, phosphatidylethanol; PNF, primary non-function; PO, oral; POD, post-operative day; PSC, primary sclerosing cholangitis; RAI, rejection activity index; re-LT, retransplantation; RFS, recurrence-free survival; RR, relative risk or risk ratio; RRT, renal replacement therapy; R₃S, Rochester Relapse Risk; rTAC, reduced tacrolimus; SALT, Sustained Alcohol Use Post-Liver Transplant; SCS, static cold storage; SMV, superior mesenteric vein; SNP, single nucleotide polymorphism; SRL, sirolimus; SRTR, Scientific Registry of Transplant Recipients; TAC, tacrolimus; TCMR, T-cell-mediated rejection; UNOS, United Network for Organ Sharing.

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transplants, or expert opinion, including the treatment of antibody-mediated rejection. Finally, further investigation is needed to truly optimize the management of the liver allograft, including the prevention of recurrent alcohol-associated, metabolic dysfunction-associated steatotic and immune-mediated liver diseases.

Keywords: Machine perfusion, living donor, alcohol liver disease, immune-mediated disease, metabolic dysfunction-associated steatohepatitis, rejection, immunosuppression

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INTRODUCTION

In 2012, the American Association for the Study of Liver Diseases (AASLD), in conjunction with the American Society of Transplantation (AST), produced a practice guideline on “Long-term management of the successful adult liver transplant.”^[1] This document summarizes the early post-transplant management of patients, technical and immunological complications, and management of immunosuppression-related complications. Since then, there have been many developments in the field of liver transplantation (LT), including the advent of direct-acting antiviral agents (DAA) to treat chronic hepatitis C (HCV), a better understanding of the different types of rejection including antibody-mediated rejection (AMR), the development of machine perfusion, and the dramatic increase in patients with alcohol-associated liver disease (ALD) and metabolic dysfunction-associated steatotic liver disease (MASLD) requiring LT. The current guideline will encompass recommendations for post-LT management in adult recipients, including the diagnosis and management of vascular and biliary complications, use of machine perfusion, clinical outcomes from utilization of donation after circulatory death (DCD) and living donors, induction immunosuppression and specific immunosuppressive regimens in different clinical settings, diagnosis and treatment of the varied types of rejection, monitoring modalities for allograft health, and specific issues regarding recurrent ALD, MASLD, and immune-mediated liver diseases.

METHODS

Literature search and grading of evidence

Literature search strategies were developed in collaboration with a medical librarian for the concepts of

immunosuppression, rejection, and recurrent disease in adult liver transplant patients. Databases searched included Ovid MEDLINE and Embase.com, and searches for gray literature were run in the Cochrane Library and relevant association websites. Initial search was completed on May 15, 2023. All searches were limited to English, humans, and from January 1, 2016, to May 15, 2023. MEDLINE and conference abstract records were excluded from the Embase search.

Studies were selected for inclusion in the systematic review of evidence if they met the following criteria: (1) peer-reviewed full-text articles; (2) population of patients with consideration for LT; (3) population of patients who have received an LT; and (4) included one of the outcomes of interest. Articles were excluded from the systematic review if they were conference abstracts that were not published in peer-reviewed journals; editorials, commentaries, case reports, narrative reviews, consensus documents, or letters; full-text articles that were not available in English; or studies that did not report on outcomes of interest.

A total of 1913 unique studies met the search term requirements (Supplemental Table S1, <http://links.lww.com/LVT/B9>). Based on a screening of titles and abstracts from these studies, 415 articles continued to full-text review and 144 articles to data extraction (Supplemental Figure S1, <http://links.lww.com/LVT/B9>).^[2] Data were not extracted from excluded articles, but these were available as discussion or background references. A refresher literature search was conducted on September 12, 2024, and relevant articles that addressed the PICO questions published since the previous search were also included. Additional articles not identified in the search but thought to be key to the evidence needed to formulate recommendations were

also included. A total of 175 studies provided data that informed the recommendations.

Guideline statements were developed and graded by members of the writing group based on available evidence, and the level of evidence was assessed using the Oxford grading system.^[3] Recommendations were then assigned the strength of recommendations based on the level of evidence, risk–benefit ratio, and patient preferences. Recommendations were graded as strong when supporting evidence was robust and consistent to allow the adoption of the recommendations by clinicians as standard of care. Recommendations were graded as weak when the evidence was weaker, inconsistent, or conditional and alternative approaches were available, to allow clinicians to be flexible in their decision-making. Agreement or disagreement on each guideline statement was voted upon by each member of the writing group using the Alchemer survey platform.

Panel composition, review process, and funding

The guideline was developed by a multidisciplinary writing group. The guideline was reviewed by the Practice Guidelines Committee, made available for public comment, approved by the AASLD Governing Board and the AST Board of Directors. The AASLD Governing Board approved the final version on July 22, 2025. Conflicts of interest were managed according to AASLD's Code for the Assessment and Management of Conflicts of Interest.^[4] This guideline was funded by the AASLD.

SURGICAL CONSIDERATIONS

Living donor liver transplantation

Question 1. Do adult living donor liver transplant recipients, compared with deceased donor liver transplant recipients, achieve superior long-term graft function and patient survival?

Guideline statements

1. Living donor liver transplantation should be considered in appropriate adult liver transplant candidates, as it has comparable long-term graft and patient survival rates when compared with deceased donor liver transplantation. (Strong, Level 2)
2. Living donor liver transplantation should be considered in appropriate adult liver transplant candidates to allow for earlier transplant at a lower MELD score with a higher intention-to-treat survival benefit. (Strong, Level 2)

3. There should be increased awareness of potential biliary and vascular complications postoperatively in adult living donor liver transplant recipients, so as to allow for early recognition and management. (Strong, Level 3)

Despite technical advances in LT, the persistence of organ shortage has necessitated innovative solutions such as living donor programs and improved organ preservation techniques. Comparison of long-term graft function, complications, and patient survival between living donor liver transplant (LDLT) and deceased donor liver transplant (DDLT) is a complex issue. Studies have shown that short-term outcomes are generally comparable between LDLT and DDLT.^[5–8] In a meta-analysis of 19 studies ($n=4571$ LDLT, $n=66,826$ DDLT), LDLT was associated with lower unadjusted mortality rates at 1, 3, and 5 years post-transplant (5-year HR 0.87, 95% CI 0.81–0.93, $p<0.0001$) and similar graft survival when compared with DDLT.^[6] Similar unadjusted short-term survival benefit from LDLT was reported by another meta-analysis of 39 studies ($n=38,563$), in which most complication rates were similar between the 2 groups except for a higher biliary complication rate (OR 2.23, 95% CI 1.59–3.13, $I^2=77\%$, $p<0.001$) and vascular complication rate (OR 2.0, 95% CI 1.31–3.07, $I^2=45\%$, $p=0.001$) in LDLT. Of note, the biliary complication rates improved with center experience.^[8] Higher rates of biliary complications and hepatic artery thromboses in LDLT were likewise observed in the Adult-to-Adult Living Donor Transplant (A2ALL) cohort study.^[9] A recent large multicenter retrospective study ($n=3633$, 18 centers) reported an incidence of 11.4% for biliary leaks and 20.6% of biliary strictures in LDLT recipients. Identified risk factors for biliary leaks included multiple bile duct anastomosis (OR 1.82, $p<0.001$), Roux-en-Y hepaticojejunostomy (OR 1.38, $p=0.04$), and a history of major abdominal surgery (OR 1.36, $p=0.03$). Risk factors for biliary anastomotic strictures included ABO incompatibility (OR 1.37, $p<0.001$), blood loss >1 L (OR 1.37, $p<0.01$), and previous abdominal surgery (OR 1.71, $p<0.001$).^[10]

Long-term outcomes are influenced by factors such as the underlying disease, recipient characteristics, and the expertise of the transplant center. A high-volume, single-center retrospective study ($n=668$ LDLT, $n=1596$ DDLT) reported a similar 20-year graft and patient survival between the LDLT and DDLT (HR 1.17, 95% CI 0.87–1.57, $p=0.29$ and HR 1.0, 95% CI 0.72–1.38, $p=0.99$, respectively) after adjustment for donor characteristics such as donor age and graft ischemic times, and patient characteristics such as lower MELD scores, higher rates of autoimmune diseases, fewer hospitalized candidates, and more candidates with malignancy in the LDLT group.^[7] While initial reports from Western countries have shown inferior LDLT outcomes in high-urgency situations,^[11]

several Asian studies have demonstrated the contrary.^[12–14] Retransplantation (re-LT), Budd–Chiari syndrome, and portal vein thrombosis (PVT) have been viewed as contraindications for LDLT in most centers. However, the increased experience and improved techniques in some Asian transplant centers show that these technical hurdles can be overcome and are not absolute contraindications for LDLT in experienced centers.^[15] In the United States, a recent analysis of the Organ Procurement and Transplantation Network (OPTN) database from 2010 to 2021 showed comparable graft survival between LDLT and DDLT recipients with MELD ≥ 25 , particularly in experienced transplant centers;^[16] however, this is tempered by another analysis of the same database from 2002 to 2021 that reported a higher risk for graft loss (aHR 1.68, $p=0.009$) for LDLT recipients with MELD ≥ 30 .^[17]

LDLT allows for earlier LT at a lower MELD score, lowering the waitlist mortality risk^[6] and decreasing waitlist drop-out for patients with HCC,^[18] which results in a lower intention-to-treat risk of death and more life-years gained.^[18,19] Regulatory requirements for LDLT are more stringent than for DDLT, and thus programs must be structured with necessary surgical, medical, and staff complements.

Liver transplantation with donation after circulatory death donors

Question 2. In adult liver transplant recipients, do donation after circulatory death livers, compared with donation after brain death livers, confer similar graft and patient survival and complication rates in adult liver transplant recipients?

Guideline statements

4. Liver transplantation in adults using donation after circulatory death organs has comparable 1-year graft and patient survival as donation after brain death livers and should be considered at experienced LT programs, particularly if machine perfusion or normothermic regional perfusion is utilized. (Strong, Level 3)
5. Donation after circulatory death livers preserved in static cold storage have a higher risk for early allograft dysfunction and ischemic cholangiopathy compared with donation after brain death livers, which can be mitigated by machine perfusion or normothermic regional perfusion. (Strong, Level 2)

Historical evidence indicates an increased risk of ischemic cholangiopathy^[20] and inferior outcomes^[21] with DCD as compared with donation after brain death

(DBD) LT. A study of the Scientific Registry for Transplant Recipients (SRTR) database ($n=33,429$, 2002–2008) reported comparable 10-year patient survival rates for DCD and DBD LT (55.5% vs. 60.2%, $p=0.24$), but the 10-year graft survival was lower in DCD than in DBD LT (45.5% vs. 56.4%, $p<0.001$), with the difference occurring mostly in the first year.^[21] A more recent study of the OPTN database ($n=32,255$ DBD, $n=2166$ DCD, 2003–2016)^[22] and a meta-analysis of 13 studies^[23] similarly observed no difference in patient survival between DCD and DBD LT, but both noted an increased risk of graft loss for DCD allografts among the heterogeneous studies (RR 1.27, 95% CI 1.16–1.38, $p=0.01$ and HR 1.50, 95% CI 1.20–1.88, $p=0.001$, $I^2=67.8\%$, respectively). More recent experience with DCD allografts utilizing a super rapid donor recovery technique^[24] produced better results, with a single-center study ($n=100$ DCD, $n=318$ DBD, 2016–2020) showing comparable 1-year graft and patient survival rates between DCD and DBD LT (93% vs. 92%, $p=0.68$ and 94% vs. 93%, $p=0.69$, respectively).^[25] The introduction of machine perfusion and normothermic regional perfusion (NRP) in recent years has further promoted DCD graft utilization, yielding comparable results as DBD grafts preserved with static cold storage (SCS).^[26,27] Accordingly, DCD use has increased from 4.7% of total LT in the United States in 2013 to 15.4% in 2023.^[28]

Complications that may be accentuated in DCD LT include ischemia-reperfusion injury, whereby DCD organs are exposed to warm ischemia during the agonal phase (time from withdrawal of life support to circulatory cessation). Similarly, studies suggest that DCD LT may have a higher incidence of biliary complications, such as ischemic-type biliary strictures. While these complications may not affect short-term survival, they can potentially impact quality of life or long-term survival. A meta-analysis of 21 studies showed a trend for more primary nonfunction (OR 1.74, 95% CI 1–3.03, $p=0.05$, $I^2=11\%$) and a higher risk for biliary complications in DCD grafts than DBD grafts (OR 2.48, 95% CI 2.05–3.0, $p<0.001$), including ischemic cholangiopathy (OR 27.73, 95% CI 16.71–46.0, $p<0.001$).^[29] Thus, donor and recipient selections are important to the success of DCD LT as is the donor recovery process, with limitations of functional warm ischemic times as well as donor hepatectomy time.^[24,30] Although there is no standard definition of extended-criteria donor livers, an International Liver Transplant Society consensus conference on DCD LT recommended careful scrutiny of overall risk factors when considering DCD livers from donors >60 years of age, body mass index >30 kg/m², with $>30\%$ steatosis, or cold ischemia time of >8 hours in the absence of machine perfusion technology and in recipients with MELD scores above 25. There was a recommendation against the use of DCD grafts for recipients with known complex PVT.^[31] Nonetheless, the

threshold for using DCD livers for various recipients continues to evolve in the era of in-situ and ex-situ perfusion.

Machine perfusion and NRP technology can mitigate the risks of early allograft dysfunction (EAD) and biliary strictures associated with DCD organs^[26,27,32–34] as discussed below, and pave the way for wider utilization of these organs. The expertise of the procurement team and the LT center plays a crucial role in achieving successful outcomes with both DCD and DBD LT.

Machine perfusion and normothermic regional perfusion

Question 3. In adult liver transplant recipients, does machine perfusion, compared with static cold storage, lead to improved short- and long-term graft and patient outcomes in adult liver transplant recipients?

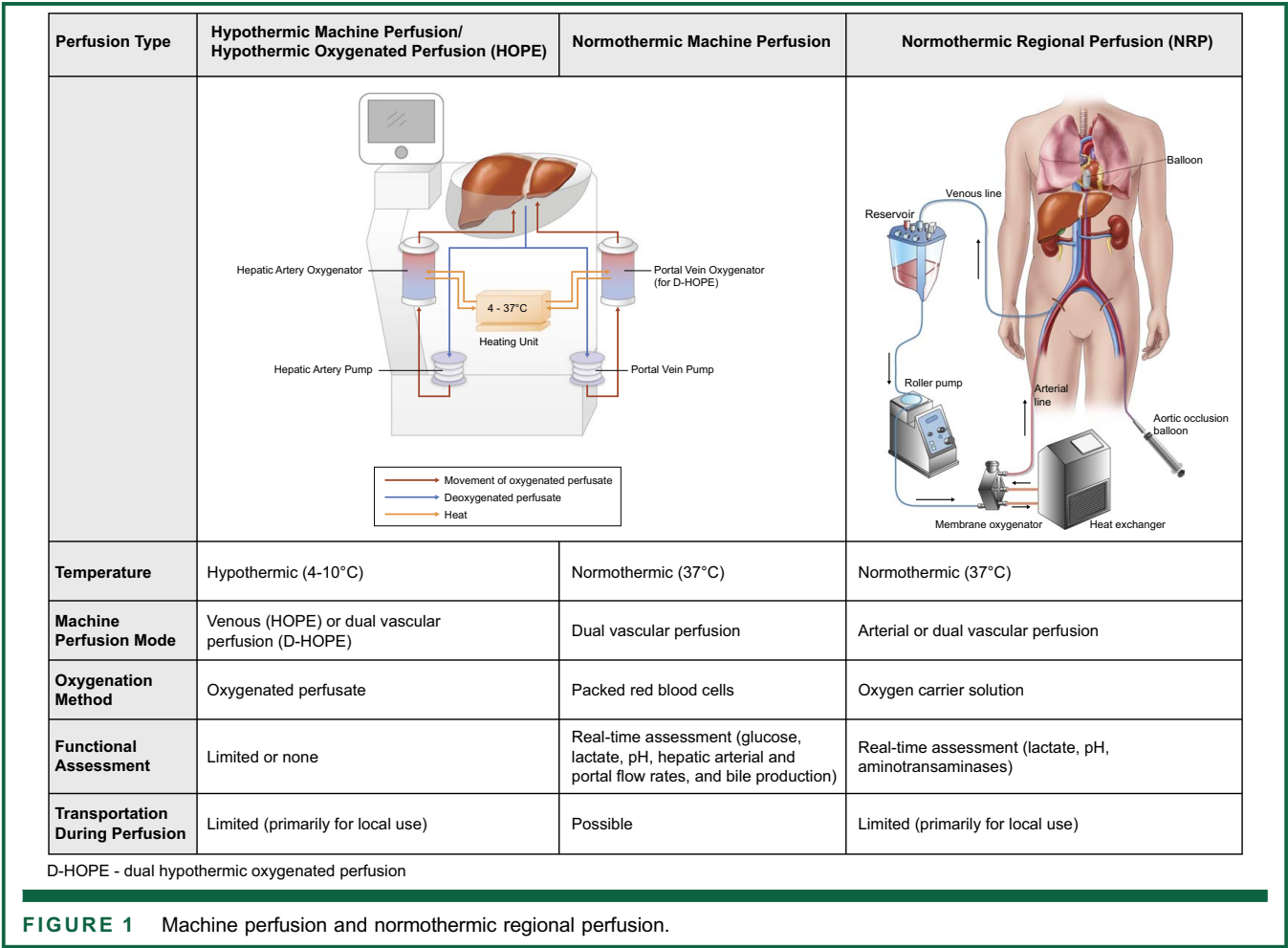
Guideline statements

6. Machine perfusion should be considered for extended-criteria donation after brain death livers and donation after circulatory death livers to decrease discard rates and increase the donor pool for adult liver transplant candidates. (Strong, Level 2)
7. Hypothermic machine perfusion and normothermic machine perfusion should be considered for extended-criteria donation after brain death livers and donation after circulatory death livers to reduce the incidence of early allograft dysfunction in adult liver transplant recipients, as compared with static cold storage. (Strong, Level 1)
8. Hypothermic machine perfusion and normothermic machine perfusion should be considered for donation after circulatory death livers to be transplanted in adult liver transplant recipients, to reduce the incidence of non-anastomotic biliary strictures, as compared with static cold storage. (Strong, Level 2)
9. Normothermic regional perfusion should be considered to reduce the incidence of early allograft dysfunction in adult liver transplant recipients of donation after circulatory death allografts. (Strong, Level 2)
10. Normothermic regional perfusion may be considered to reduce non-anastomotic biliary strictures in adult liver transplant recipients of donation after circulatory death allografts. (Weak, Level 3)

Machine perfusion (Figure 1) refers to the use of a mechanical device to support and preserve a donor liver before transplantation, which can be accomplished via ex-situ hypothermic machine perfusion (HMP), also known as hypothermic oxygenated machine perfusion (HOPE) or dual-HOPE (D-HOPE), or ex-situ normothermic machine perfusion (NMP). HMP uses a highly oxygenated artificial solution at hypothermic temperature (8–12 °C) and can be applied for 1–2 hours shortly before implantation. In contrast, NMP uses a blood-based perfusate at 37 °C and can be applied immediately after procurement at the donor site or after graft transport with SCS for a longer duration of 3–12 hours,^[33] although long-term use up to 7 days has also been reported.^[35] In controlled DCD donors, in situ postmortem NRP can be used to restore perfusion to organs for 2–4 hours with warm oxygenated blood after circulatory cessation has been declared, while procurement is taking place. In thoracoabdominal NRP, cannulas are placed into the right atrium and the descending aorta, and the brachiocephalic arteries are clamped to occlude blood flow to the brain, followed by re-initiation of cardiac activity. Abdominal NRP limits perfusion only to abdominal organs via the infrarenal aorta and IVC (or alternatively the femoral artery and vein) and does not involve the re-initiation of cardiac activity.^[36] An extracorporeal membrane oxygenation machine is then used to provide warm perfusion to all the organs, restoring their function before procurement. Recovered organs are then preserved with SCS or machine perfusion. NRP has become routinely used in current DCD procurement in Spain, France, and Italy.^[37]

Some of the best applications of machine perfusion in LT include graft preservation from extended-criteria donors^[38] and optimization of DCD allografts.^[39] Machine perfusion may also potentially recondition marginal or steatotic donor livers and mitigate ischemia-reperfusion injury to improve the overall function of the liver.^[40] Both machine perfusion types consistently improved EAD rates in 5 meta-analyses^[26,27,32,33,41] (Table 1). In most studies, it also led to decreased non-anastomotic biliary stricture rates,^[27,32,33,41] even more so in DCD grafts.^[26,27] Acute cellular rejection episodes were also reduced with machine perfusion as compared with SCS (OR 0.55, 95% CI 0.33–0.91, $p=0.02$ and RR 0.54, 95% CI 0.31–0.94, $p=0.03$).^[41,42] One-year graft^[26,27,33,41] and patient survival^[26,27] were also improved with HMP, but there are no data on long-term survival.

DCD livers may benefit the most from machine perfusion and NRP. In a randomized controlled trial (RCT) ($n=160$), non-anastomotic biliary strictures occurred in fewer DCD livers treated with machine perfusion than in those preserved with SCS (RR 0.36, 95% CI 0.14–0.94, $p=0.03$).^[43] A large multicenter



retrospective cohort study (n=242) also demonstrated less EAD rates (36.4% vs. 56.1%, $p < 0.01$), biliary anastomotic strictures (6.7% vs. 22.4%, $p = 0.001$), and ischemic cholangiopathy (0% vs. 9%, $p < 0.01$) in DCD livers recovered with NRP as compared with standard super rapid recovery, although patient and graft survival were comparable.^[34] European countries routinely using machine perfusion or NRP had a higher DCD liver utilization rate (64.9%) as compared with countries with selective use (28.4%).^[44] The field of machine perfusion and NRP continues to rapidly evolve, and its use is approaching the standard of care while data on patient prioritization, resource utilization, and economics of use are awaited. A real-world analysis demonstrated that despite a higher cost of organ acquisition and preservation with the use of NMP for both extended-criteria DBD and DCD liver grafts, its overall 90-day health care costs-per-transplantation were comparable to those using SCS.^[45] Access to NRP remains limited at this time, but additional evidence is anticipated in the near future that may further strengthen the rationale for expansion of its use.

Early allograft dysfunction

Question 4. Are pharmacological interventions effective in preventing early allograft dysfunction in adult liver transplant recipients?

- Guideline statements
11. Continuous risk score models incorporating dynamic changes in post-transplant liver function tests, for example, the Liver Graft Assessment Following Transplantation (L-GrAFT) risk model score or Early Allograft Failure Simplified Estimation (EASE) score, should be used to predict the outcome following early allograft dysfunction in adult liver transplant recipients. (Strong, Level 2)

12. Pharmacologic interventions (eg, N-acetylcysteine, prostaglandins) cannot be routinely recommended to prevent early allograft dysfunction or to improve clinical outcomes in adult liver transplant recipients. (Weak, Level 2)

TABLE 1 Summary of clinical outcomes with machine perfusion and normothermic regional perfusion from meta-analyses

	Ramirez-Del Val et al, 2022 ^[32]	Tingle et al, 2023 ^[26] (Cochrane Review)	Liang et al, 2023 ^[27]	Parente et al, 2023 ^[33]	Tang et al, 2024 ^[41]
Study arms	1. HMP vs. SCS (12 studies) 2. NMP vs. SCS (13 studies) 3. NRP vs. SCS (13 studies)	1. HMP vs. SCS (4 RCTs) 2. NMP vs. SCS (3 RCTs)	1. HMP vs. SCS (14 studies) 2. NMP vs. SCS (16 studies) 3. NRP vs. SCS (16 studies)	1. HMP vs. SCS (4 RCTs) 2. NMP vs. SCS (3 RCTs)	HMP vs. SCS (5 RCTs +6 cohort studies)
Outcomes					
Post-reperfusion syndrome	1. OR 0.33, 95% CI 0.15–0.74, $p < 0.01$ 2. OR 0.51, 95% CI 0.29–0.9, $p = 0.02$				RR 0.92, 95% CI 0.57–1.48, $p = 0.08$
EAD	1. OR 0.51, 95% CI 0.35–0.75, $p < 0.001$ 2. OR 0.66, 95% CI 0.45–0.97, $p = 0.03$ 3. OR 0.52, 95% CI 0.38–0.7, $p < 0.001$	1. OR 0.35 (95% CI 0.23–0.53) 2. OR 0.40 (95% CI 0.22–0.74)	1. OR 0.39, 95% CI 0.3–0.53, $p < 0.01$ 2. OR 0.55, 95% CI 0.34–0.88, $p = 0.01$ 3. OR 0.69, 95% CI 0.54–0.9, $p < 0.01$	1. RR 0.48, 95% CI 0.35–0.65, $p < 0.001$ 2. RR 0.5, 95% CI 0.3–0.86, $p = 0.01$	RR 0.54, 95% CI 0.42–0.68, $p < 0.001$
Median ICU LOS	1. –0.21 d, 95% CI –1.71 to 1.27, $p = 0.77$ 2. –7.07 d, 95% CI –8.76 to –5.38, $p < 0.001$				0.19 d, 95% CI –0.19 to 0.56, $p = 0.33$
Median hospital LOS	1. –3.9 d, 95% CI –5.92 to –1.89, $p < 0.001$ 2. –12.41 d, 95% CI –17.36 to –7.46, $p < 0.001$				–1.38 d, 95% CI –2.87 to 0.12, $p = 0.07$
Peak ALT	Median difference 1. –314.19, 95% CI –492.38 to –35.99, $p < 0.001$ 2. –412.17, 95% CI –595.38 to –228.97, $p < 0.001$			Mean difference 1. –253.32, 95% CI –425.64 to –80.99, $p = 0.004$	
Peak AST	Median difference 2. –453.9, 95% CI –686.78 to –221.01, $p < 0.001$			Mean difference 1. –484.53, 95% CI –808.07 to –160.98, $p = 0.003$ 2. –21.09, 95% CI –341.56 to 299.38, $p = 0.9$	
Primary nonfunction	1. OR 0.73, 95% CI 0.26–2.03, $p = 0.75$ 2. OR 1.3, 95% CI 0.16–10.56, $p = 0.56$ 3. OR 0.51, 95% CI 0.27–0.98, $p = 0.04$			1. RR 0.33, 95% CI 0.08–1.44, $p = 0.14$ 2. RR 2.51, 95% CI 0.10–60.91, $p = 0.57$	RR 0.54, 95% CI 0.22–1.30, $p = 0.76$
Postoperative RRT	1. OR 0.88, 95% CI 0.5–1.56, $p = 0.67$ 2. OR 1.16, 95% CI 0.61–2.23, $p = 0.92$				RR 0.98, 95% CI 0.64–1.51, $p = 0.43$

1-year graft loss (as HR, RR) or survival (as OR)	1. HR 0.45 (95% CI 0.23–0.87) 2. HR 1.2 (95% CI 0.44–3.29)	1. OR 2.31, 95% CI 1.54–3.45, $p < 0.01$ 2. OR 1.11, 95% CI 0.75–1.65, $p = 0.6$ 3. OR 1.44, 95% CI 0.64–3.25, $p = 0.38$	1. RR 0.40, 95% CI 0.17–0.95, $p = 0.04$ 2. RR 1.42, 95% CI 0.52–3.88, $p = 0.49$	RR 0.38, 95% CI 0.21–0.68, $p = 0.001$
1-year patient loss (as HR, RR) or survival (as OR)	1. HR 0.91 (95% CI 0.42–1.98) 2. HR 1.08 (95% CI 0.31–3.8)	1. OR 1.79, 95% CI 1.15–2.79, $p = 0.01$ 2. OR 1.13, 95% CI 0.74–1.71, $p = 0.57$ 3. OR 1.15, 95% CI 0.53–2.46, $p = 0.73$		RR 0.67, 95% CI 0.31–1.45, $p = 0.31$
Re-LT			1. RR 0.21, 95% CI 0.04–0.96, $p = 0.04$ 2. RR 1.33, 95% CI 0.58–3.04, $p = 0.49$	RR 0.42, 95% CI 0.19–0.93, $p = 0.03$
Overall biliary complications			1. RR 0.78, 95% CI 0.6–1.02, $p = 0.07$ 2. RR 0.68, 95% CI 0.46–1.02, $p = 0.06$	RR 0.74, 95% CI 0.62–0.91, $p = 0.004$
Ischemic biliary complications or NABS	1. OR 0.32 (95% CI 0.12–0.83) 2. OR 0.78 (95% CI 0.27–2.27)	1. OR 0.43, 95% CI 0.26–0.7, $p < 0.01$ 2. OR 0.55, 95% CI 0.36–0.84, $p < 0.01$ 3. OR 0.97, 95% CI 0.17–5.46, $p = 0.98$	1. RR 0.43, 95% CI 0.2–0.93, $p = 0.03$ 2. RR 0.46, 95% CI 0.17–1.22, $p = 0.12$	RR 0.34, 95% CI 0.19–0.61, $p < 0.001$
Major complications		Clavien–Dindo Grade ≥ 3 1. OR 0.62, 95% CI 0.46–0.84, $p < 0.01$ 2. OR 0.65, 95% CI 0.32–1.31, $p = 0.23$ 3. OR 0.56, 95% CI 0.06–4.93, $p = 0.6$	Clavien $\geq 3b$ 1. RR 0.76, 95% CI 0.63–0.93, $p < 0.01$	Clavien–Dindo Grade ≥ 3 RR 0.80, 95% CI 0.62–1.04, $p = 0.04$

Note: This is a rapidly evolving field, and new data may be available at the time of publication of this guideline.

Abbreviations: CI, confidence interval; EAD, early allograft dysfunction; HMP, hypothermic machine perfusion; HR, hazard ratio; ICU, intensive care unit; LOS, length of stay; NABS, non-anastomotic biliary strictures; NMP, normothermic machine perfusion; NRP, normothermic regional perfusion; re-LT, retransplantation; OR, odds ratio; RR, relative risk or risk ratio; RCT, randomized controlled trial; RRT, renal replacement therapy; SCS, static cold storage.

EAD is the clinical phenotype of a poorly functioning liver allograft, which can be caused by a multitude of donor, recipient, and perioperative factors. EAD has been associated with inferior short-term graft and patient survival outcomes, as well as perioperative complications and morbidity. The most commonly used definition of EAD by Olthoff et al^[46] is a binary characterization of graft function based on the presence of peak AST/ALT higher than 2000 U/L within the first 7 days, serum bilirubin of ≥ 10 mg/dL, or INR ≥ 1.6 on postoperative day 7. However, this binary characterization fails to capture the continuum over which graft dysfunction exists.

In recent years, dynamic models (eg, kinetic models) have been developed, accounting for both the degree of laboratory abnormalities and their rate of normalization. In 2017, the L-GrAFT score was created based on >2000 primary adult LT recipients from a single center.^[47] Calculated from post-LT day 10 (L-GrAFT₁₀) or day 7 (L-GrAFT₇) results for bilirubin, INR, and platelets, the L-GrAFT score was far superior in discriminating 3-month graft failure-free survival compared with another model, the Model for Early Allograft Function (MEAF) score, and the binary EAD definition (C-statistics 0.85 vs. 0.70 vs. 0.68, respectively).^[48] Subsequently, a large-scale multicenter validation of L-GrAFT score on >3200 patients from 3 independent centers confirmed the superiority of L-GrAFT₁₀ and L-GrAFT₇ scores over the Olthoff criteria and MEAF score (C-statistics 0.82 and 0.78 vs. 0.68 and 0.72, respectively). Furthermore, in a post hoc analysis of the European normothermic machine perfusion (COPE NMP) trial, the L-GrAFT₇ score not only maintained excellent discrimination in predicting 3-month graft failure-free survival compared with the Olthoff criteria and MEAF score (C-statistics 0.81 vs. 0.64 and 0.57, respectively), but was also significantly associated with time to liver allograft complications, Clavien $\geq 3b$ complications (ie, intervention under general anesthesia, life-threatening complication, organ dysfunction, or death), post-LT renal replacement therapy, and length of hospitalization.^[49]

Subsequently, a modified kinetic model, the Early Allograft Failure Simplified Estimation (EASE) score, was developed from an Italian multicenter cohort (n=1609).^[50] In addition to dynamic parameters of AST, bilirubin, and platelets, the model also included recipient MELD score, transfusion of packed red blood cells, early thrombosis of hepatic vessels, and center volume to accurately predict 3-month graft failure-free survival and need for re-LT. L-GrAFT and EASE scores have demonstrated comparable accuracies in discriminating 3-month graft failure-free survival in external validation datasets that are independent from derivation datasets, and consistently outperform the Olthoff criteria and MEAF score in prognosticating post-LT outcomes. Multicenter international data support the use of L-GrAFT or EASE scores as accurate models to characterize early allograft function following LT.

EAD significantly impacts not only graft and patient outcomes, but perioperative complications and morbidity as well. As such, there has been significant interest in mitigating the impact of EAD following LT. The medical interventions most reported include IV N-acetylcysteine (NAC) and prostaglandin therapy, although data supporting their use remain limited. An RCT (n=150 LDLT recipients) showed lower AST and ALT levels in recipients receiving IV NAC, but no difference in the composite primary endpoint of EAD and acute kidney injury.^[51] In the NAC-400 single-center RCT (n=214), there was no difference in the primary endpoint of EAD between use of NAC or standard-of-care during liver procurement (31% vs. 37.4%, $p=0.38$).^[52] A more recent meta-analysis of 13 studies (n=1121, in which 550 received NAC) revealed a reduced incidence of primary graft nonfunction, postoperative complications, and peak AST and ALT.^[53] However, significant heterogeneity in the various modes of delivery of NAC (donor or recipient) makes interpretation and generalization of these findings difficult. A recent Cochrane review (n=11 RCT, 771 patients) evaluated the harms and benefits of prostaglandins in adult patients undergoing LT.^[54] While prostaglandins reduced the risk for acute kidney failure requiring dialysis [relative risk (RR) 0.42, 95% CI 0.24–0.73, $p<0.01$], there was no significant difference in all-cause mortality at 1 month (RR 0.86, 95% CI 0.61–1.23, $p=0.41$), serious adverse events (RR 0.92, 95% CI 0.60–1.40, $p=0.68$), re-LT (RR 0.98, 95% CI 0.49–1.96, $p=0.95$), EAD (RR 0.62, 95% CI 0.33–1.18, $p=0.15$), or primary graft nonfunction (RR 0.58, 95% CI 0.26–1.32, $p=0.19$).

Technical graft complications

Question 5. In adult liver transplant recipients with vascular complications, when is surgical correction, percutaneous intervention, or liver retransplantation the most appropriate management strategy?

Guideline statements

13. In adult liver transplant recipients, hepatic artery or portal vein stenosis or thrombosis should be managed in a timely fashion to prevent graft loss, which is particularly urgent in the setting of hepatic artery involvement. Surgical or percutaneous endovascular intervention and anticoagulation should be considered for cases detected in the immediate postoperative period, while percutaneous endovascular interventions and/or anticoagulation should be considered for diagnoses made later. (Strong, Level 4)

14. In adult liver transplant recipients, timely retransplantation should be considered when vascular complications result in severe allograft injury or allograft failure. (Strong, Level 2)

While LT is a life-saving procedure for patients suffering from end-stage liver disease, it is a technically challenging operation and can be accompanied by a spectrum of technical complications involving the hepatic artery, portal vein, hepatic veins, or inferior vena cava. Such complications can, in turn, significantly impact graft and patient outcomes. Thus, early recognition and management of these complications are crucial in preserving allograft health (Table 2). The diagnosis of vascular complications after LT is based on radiologic imaging such as Doppler ultrasound, CT, magnetic resonance (MR) angiography or venography, or catheter angiography.^[55–60] Doppler ultrasound is widely used as a screening tool to assess liver graft vasculature in the immediate postoperative period, due to its noninvasive nature, lack of contrast exposure, accessibility, and lower cost. Most cases of hepatic artery thrombosis (HAT) are detected within 5 days after LT, while PVT and hepatic venous outflow obstruction are usually diagnosed later.^[61] Most centers obtain at least 1 Doppler ultrasound within the first 5 days after LT, although practice varies widely from center to center, and the optimal schedule remains to be established.^[61–63] Contrast-enhanced CT angiography or venography provides more detail when vascular abnormality is suspected or when Doppler ultrasound findings are inconclusive. MRI can be used as an alternative test when there is a contraindication to a contrast-enhanced CT scan.

Early HAT within 5 days of LT is usually corrected surgically or endovascularly, with anticoagulation, depending on the patient's condition and the transplant center's practice.^[64–66] Endovascular management can confer some risk of hepatic artery injury and rupture in the very early postoperative period, but it can be the first-line therapy when the patient's condition precludes surgery. When early HAT leads to progressive liver failure, urgent re-LT is needed to avoid a high mortality risk.^[67] If HAT is identified within 7 days of LT, patients are prioritized as status 1 on the transplant waitlist.^[68] Late-onset HAT can be addressed with percutaneous interventions and/or anticoagulation alone, but if it occurs gradually, adequate collateralization may develop that allows the graft to survive without intervention.^[59] However, ischemic cholangiopathy that may require re-LT in the future may still develop in this setting.^[69] Similarly, early PVT or stenosis can also be addressed surgically or endovascularly with anticoagulation,^[66] and later cases are managed endovascularly with or without anticoagulation, or with anticoagulation alone. The choice of intervention is

mainly driven by center practice and the condition of the patient.^[59]

Question 6. In adult liver transplant recipients with anastomotic or non-anastomotic biliary strictures, does magnetic resonance cholangiopancreatography (MRCP) provide comparable diagnostic accuracy to endoscopic retrograde cholangiopancreatography (ERCP)?

Guideline statements

15. MRCP should be used as the first-line diagnostic tool for anastomotic and non-anastomotic biliary strictures in adult liver transplant recipients. (Strong, Level 2)
16. In adult liver transplant recipients in whom there is a high clinical suspicion for biliary pathology or for the need of endoscopic intervention, an ERCP can be pursued, without a preceding MRCP or following a negative MRCP. (Weak, Level 5)

Biliary strictures following LT occur in up to 15% of DDLT recipients and 40% of LDLT recipients, significantly impacting morbidity and graft survival.^[70] Biliary strictures can be either anastomotic or non-anastomotic. Anastomotic biliary strictures are typically single, short-segment strictures located at the site of the biliary anastomosis (choledochocholedochostomy, choledochoduodenostomy, Roux-en-Y choledochojejunostomy).^[71,72] Non-anastomotic biliary strictures can be multiple and involve the biliary tree proximal to the anastomosis, often due to perioperative ischemic insults to the peri-biliary vasculature from long cold or warm preservation time or hepatic artery insufficiency or thrombosis.

The presentation of post-LT biliary strictures ranges from asymptomatic elevation of liver chemistry tests to overt biliary obstruction with jaundice and/or cholangitis. MRCP has a 94%–96% sensitivity and a 94%–95% specificity for detecting biliary strictures and allows for a noninvasive evaluation of the intrahepatic and extrahepatic biliary tree as well as for identification of bilomas or intraductal biliary casts.^[70] However, when the pretest probability of a biliary stricture is high and the need for intervention is likely, proceeding directly to ERCP is reasonable.^[73] ERCP can also be pursued despite a negative MRCP when there is high clinical suspicion for a biliary pathology.

Question 7. In adult liver transplant recipients with anastomotic and non-anastomotic biliary strictures, does endoscopic intervention provide better outcomes than percutaneous transhepatic biliary drain placement?

TABLE 2 Vascular complications after liver transplantation^[55–60]

Complication	Incidence	Risk factors	Presentation	Diagnosis	Treatment	Long-term sequelae
Hepatic artery thrombosis	2%–10%	Hepatic artery dissection, splenic artery steal syndrome, decreased luminal diameter, variant anatomy requiring back-table reconstruction, and graft injury due to ischemic reperfusion injury	Elevated liver enzymes, graft dysfunction, and bile leak	Doppler ultrasound, CTA, MRA, and angiography	Early: Surgical thrombectomy and revascularization with anticoagulation, and re-LT. Late: IR thrombolysis, possible stenting and/or anticoagulation, and re-LT	Biliary complications (bile leak, biliary stricture, ischemic cholangiopathy) and graft loss requiring re-LT
Hepatic artery stenosis	2%–9%	Hepatic artery dissection, technical factors	Elevated liver enzymes	Doppler ultrasound, CTA, MRA, and angiography	Early: surgical revision with anticoagulation. Late: anticoagulation, percutaneous transluminal angioplasty, and stenting	Biliary complications (bile leak, biliary stricture, ischemic cholangiopathy)
Portal vein (PV) thrombosis	1%–2.6%	Technical factors (kinking, redundancy), recipient PV thrombosis, large portosystemic shunts, hypercoagulable state, extra-anatomic PV reconstruction (jump graft from SMV/IMV/ coronary)	Elevated liver enzymes, prolonged prothrombin time, graft dysfunction, and ascites	Doppler ultrasound, CTV, and MRV	Early: thrombectomy with surgical revision and anticoagulation Late: anticoagulation, percutaneous thrombolysis, stent, and TIPS	Portal hypertension (ascites, varices) and graft dysfunction
PV stenosis	2%–3%	Early (technical factors) Late (neo-intimal hyperplasia, scarring)	Ascites and elevated liver tests	Doppler ultrasound, CTV, and MRV	Percutaneous transluminal angioplasty and stenting with anticoagulation	Portal hypertension (ascites, varices) and graft dysfunction
Hepatic venous or inferior vena cava outflow obstruction (stenosis/thrombosis)	< 5%	Technical factors (piggy-back > bi-caval technique)	Ascites, hepatomegaly, renal dysfunction, portal hypertension, lower extremity edema, and graft dysfunction	Doppler ultrasound, CTV, and MRV	Angioplasty, stenting with anticoagulation, and surgical revision	Graft dysfunction and hepatic congestion

Abbreviations: CTA, computed tomographic angiography; CTV, computed tomographic venography; IMV, inferior mesenteric vein; IR, interventional radiology; MR, magnetic resonance; MRA, magnetic resonance angiography; MRV, magnetic resonance venography; re-LT, retransplantation; SMV, superior mesenteric vein.

Guideline statements

17. Endoscopic characterization, balloon dilation, and stenting should be performed as first-line management of anastomotic biliary strictures in adult liver transplant recipients who have a choledochocholedochostomy (duct-to-duct anastomosis) or choledochoduodenostomy. (Strong, Level 2)
18. Either multiple plastic stents or covered self-expandable metal stents should be used to manage anastomotic strictures endoscopically in adult liver transplant recipients, as they have similar efficacy in achieving resolution; covered self-expandable metal stents, however, may allow for fewer overall procedures and a lower cost. (Strong, Level 2)
19. Percutaneous transhepatic biliary drains should be reserved for adult liver transplant recipients with biliary strictures that have failed endoscopic management. (Strong, Level 3)
20. Either endoscopic stents or percutaneous transhepatic biliary drains should be placed for anastomotic biliary strictures in adult liver transplant recipients having a Roux-en-Y choledochojejunostomy, depending on the center's expertise. (Strong, Level 3)
21. In adult liver transplant recipients with anastomotic biliary strictures that are refractory to endoscopic and/or percutaneous transhepatic biliary drain placement, surgical revision should be considered. (Strong, Level 4)
22. Initial management of non-anastomotic biliary strictures in adult liver transplant recipients should be via endoscopic retrograde cholangiography and stenting, if the locations are amenable to such intervention. (Strong, Level 4)
23. In adult liver transplant recipients with non-anastomotic biliary strictures that failed and/or progressed despite multiple endoscopic interventions, percutaneous transhepatic biliary drain placement can be considered. (Weak, Level 4)
24. In adult liver transplant recipients with irreversible symptomatic biliary disease or progressive ischemic cholangiopathy, retransplantation should be considered. (Strong, Level 4)

Endoscopic management of anastomotic biliary strictures with sphincterotomy, balloon dilation, and stent placement is the preferred therapeutic approach in recipients with a duct-to-duct anastomosis. However, percutaneous transhepatic biliary drain placement may be appropriate in patients with surgically altered foregut anatomy, such as a Roux-en-Y gastrojejunostomy or who are critically ill with hemodynamic instability.^[73] The use of either multiple plastic stents or covered self-expandable metal stents has high reported rates of technical success and resolution of anastomotic biliary strictures across numerous case series.^[74–76] Two recent systematic reviews and meta-analyses comparing multiple plastic stents and covered self-expandable metal stents showed no difference in anastomotic biliary stricture resolution [risk ratio (RR) 0.89, 95% CI 0.4–2.02, $p=0.79$ and OR 0.99, 95% CI 0.48–2.01, $p=0.97$, respectively] or recurrence [relative risk (RR) 0.55, 95% CI 0.22–1.38, $p=0.2$ and OR 1.71, 95% CI 0.87–3.38, $p=0.12$, respectively].^[77,78] However, covered self-expandable metal stents were associated with fewer procedures (mean difference 1.69 ERCPs, 95% CI 1.0–2.39, $p<0.001$) and a shorter treatment period to stricture resolution (mean difference 40.2 days, 95% CI 3.9–76.4, $p=0.03$).^[77] The cost for stricture resolution has been demonstrated to be 25% less with self-expandable metal stents than with multiple plastic stents.^[75,78]

Drainage via the percutaneous transhepatic route is typically reserved for patients with complex strictures, bile ducts unable to be traversed by ERCP, or in LT recipients with Roux-en-Y reconstructions that make endoscopic access to the biliary tree challenging.^[70,79] In recipients with non-anastomotic biliary stricture and/or intrahepatic bilomas or biliary casts, a percutaneous biliary drain is often required. Surgical revision is typically the last resort for anastomotic biliary stricture refractory to endoscopic or percutaneous interventions,^[80–83] with a limited role in patients with multiple or proximal non-anastomotic biliary strictures. Persistent or progressive biliary strictures despite multiple interventions or the development of diffuse ischemic cholangiopathy can lead to complications such as cholangitis, hepatic abscesses, and graft dysfunction, and hence, may require re-LT.^[69,84] Given that DCD grafts confer a higher risk of ischemic cholangiopathy, increased priority on the liver waitlist is awarded to recipients who received DCD grafts and have 2 of the following: persistent cholestasis with serum bilirubin >2 mg/dL, recurrent cholangitis with associated bacteremia, or non-anastomotic biliary strictures unresponsive to treatment.^[85]

IMMUNOSUPPRESSION

Induction immunosuppression

Question 8. In adult liver transplant recipients, does induction immunosuppression provide clinical benefit and improve overall graft and patient survival?

Guideline statements

25. In adult liver transplant recipients, induction immunosuppressive therapy with basiliximab can be used to delay the start of calcineurin inhibitors to improve renal recovery in both deceased donor and living donor liver transplant recipients with renal dysfunction. (Weak, Level 2)
26. In adult liver transplant recipients, anti-thymocyte globulin should not be used as induction immunosuppressive therapy, as it has a higher mortality and graft failure risk compared with corticosteroids. (Strong, Level 2)
27. Due to insufficient data, routine use of basiliximab as induction immunosuppressive therapy is not recommended in simultaneous liver–kidney or ABO-incompatible adult liver transplants. (Strong, Level 5)

Most LT programs use parenteral corticosteroids in the immediate postoperative phase. In addition, induction immunosuppressive therapy, defined as potent immunosuppression given immediately before liver transplantation, is currently used in 27% of all LT recipients to minimize episodes of rejection within the first 3 months of LT.^[86] It may also allow for minimization of calcineurin inhibitors (CNI) and corticosteroid use to decrease their adverse effects. Induction immunosuppressive therapies used in LT recipients consist of lymphocyte-depleting induction agents (ie, anti-thymocyte globulin, anti-lymphocyte globulin, and alemtuzumab) as well as non-lymphocyte-depleting induction agents or interleukin-2 receptor antagonists, (IL2-RA) (ie, basiliximab and daclizumab). However, both anti-lymphocyte globulin and daclizumab are no longer commercially available. Alemtuzumab has been associated with a higher risk for infection and malignancy, but recent data on basiliximab and anti-thymocyte globulin did not demonstrate this risk.^[87]

Anti-thymocyte globulin, an infusion of polyclonal rabbit-derived antibodies against human T cells, has been used as an induction therapy and to treat rejection since the 1980s. However, its benefit in minimizing acute rejection at 1-year post-LT remains uncertain in the 6 RCTs available.^[88–93] Results from alemtuzumab, a human monoclonal antibody against CD52 located on T and B lymphocytes, are even less favorable.^[94] These results do not support their use in LT recipients.

Basiliximab is currently the only commercially available IL2-RA and functions by selectively inhibiting T lymphocyte proliferation. In a Cochrane analysis

evaluating induction immunosuppression with corticosteroids, anti-thymocyte globulin, basiliximab, or daclizumab in adults undergoing LT, only basiliximab, with or without corticosteroids, was found to reduce all-cause mortality (HR 0.53, 95% CI 0.31–0.93) and graft failure (HR 0.44; 95% CI 0.28–0.70) compared with corticosteroids alone. However, the evidence is based on 2 small RCTs at high risk of bias.^[95–97] A 2005–2018 analysis of the United Network for Organ Sharing (UNOS) database ($n=69,349$) showed that induction therapy with either IL2-RA [adjusted HR (aHR) 0.90, 95% CI 0.86–0.95, $p<0.001$] or lymphocyte-depleting agents (aHR 0.91, 95% CI 0.85–0.97, $p<0.001$) conferred a lower risk for a combined outcome of graft failure and/or death compared with no induction therapy. In addition, IL2-RA were associated with an improved glomerular filtration rate (GFR) (by 3.8 mL/min per 1.73 m², $p<0.001$) at 6 months after LT in patients not on dialysis and a 12% reduction in biopsy-proven acute rejection (BPAR) in the first year (adjusted OR 0.88, 95% CI 0.81–0.96).^[86] Several RCT have similarly shown improved renal function with IL2-RA use by allowing reduction and/or delay in tacrolimus (TAC) dosing and lower BPAR rates at 6 months,^[98–102] as well as lower rates of de novo diabetes mellitus (DM) and hypertension in LDLT by allowing avoidance or reduction of steroid dosing^[96,99] (Table 3). Thus, available data favor the use of IL2-RA induction in LT recipients at high risk for renal dysfunction. However, induction therapy does increase costs.^[105]

Special populations may have unique immunologic challenges that need to be addressed, including simultaneous liver, kidney, and ABO-incompatible transplants. An analysis of simultaneous liver–kidney transplant cases in the UNOS database ($n=5172$, 1996–2016) showed 5-year patient survival at 68% for those who received lymphocyte-depleting induction, 74% in the IL2-RA group, and 71% in those with no induction therapy ($p<0.001$). However, multivariate analysis showed the type of induction had no impact on patient survival.^[106] The use of ABO-incompatible LT has been limited by severe T-cell-mediated rejection (TCMR), AMR, hyperacute rejection, HAT, hepatic necrosis, and biliary issues. Strategies are still being developed to reduce preformed anti-donor ABO antibodies and to formulate desensitization protocols with rituximab and plasmapheresis before and immediately after LT. There are currently no conclusive data to support the use of induction immunosuppressive therapy in this population.

Maintenance immunosuppression and graft monitoring

Question 9. In adult liver transplant recipients, does steroid-free immunosuppression, compared with

[illegible]

Abbreviations: AZA, azathioprine; BAS, basiliximab; BPAR, biopsy-proven acute rejection; CSA, cyclosporine; DAC, daclizumab; dTAC, delayed tacrolimus; GFR, glomerular filtration rate; HTN, hypertension; IL2-RA, interleukin-2 receptor antagonists; LDLT, living donor liver transplant; MMF, mycophenolate mofetil; MDRD, modification of diet in renal disease; NS, not statistically significant; rTAC, reduced tacrolimus; r/d, reduced or delayed; TAC, tacrolimus.

steroid-based immunosuppression, confer similar graft and patient survival?

Guideline statements

28. In adult liver transplant recipients, steroid-free immunosuppression cannot be routinely recommended due to higher rates of acute rejection, steroid-resistant rejection, and decreased glomerular filtration rate when compared with steroid-based immunosuppression, despite similar graft and patient survival rates. (Strong, Level 2)

Due to the multiple side effects from corticosteroids, including hypertension, dyslipidemia, obesity, DM, osteoporosis, and infections, steroid-sparing or steroid-free immunosuppressive regimens have long been considered. A Cochrane meta-analysis of 16 RCTs ($n=1347$) has assessed the efficacy and safety of corticosteroid minimization in TAC-based immunosuppressive regimens compared with maintenance corticosteroids. It showed uncertainty of benefit or harm for corticosteroid avoidance or withdrawal versus corticosteroid continuation on the primary outcomes of patient mortality (RR 1.15, 95% CI 0.93–1.44), graft loss (RR 1.15, 95% CI 0.90–1.46), or infection (RR 0.88, 95% CI 0.73–1.05).^[107] There were 9 RCTs ($n=782$) comparing postoperative corticosteroid avoidance (excluding intraoperative corticosteroids) to corticosteroid-containing regimens and 6 RCTs ($n=565$) comparing corticosteroid withdrawal after 3–6 months to long-term corticosteroid use. Neither corticosteroid avoidance or withdrawal had significant impact on mortality (RR 1.11, 95% CI 0.84–1.48, $p=0.46$, $I^2=0\%$, and RR 1.22, 95% CI 0.86–1.72, $p=0.27$, $I^2=34\%$, respectively) nor on graft loss (RR 1.03, 95% CI 0.7–1.39, $p=0.82$, $I^2=0\%$, and RR 1.44, 95% CI 0.93–2.24, $p=0.1$, $I^2=55\%$, respectively). Corticosteroid avoidance or withdrawal was associated with less hypertension (RR 0.76, 95% CI 0.65–0.9, $p=0$, $I^2=52\%$) and DM (RR 0.81, 95% CI 0.66–0.99, $p=0.04$, $I^2=34\%$), but not hyperlipidemia (RR 0.75, 95% CI 0.38–0.1.48, $p=0.4$, $I^2=0\%$). These benefits were offset by higher rates of BPAR (RR 1.33, 95% CI 1.08–1.64, $p=0.01$, $I^2=0\%$), steroid-resistant rejection (RR 2.14, 95% CI 1.13–4.02, $p=0.02$, $I^2=0\%$), and increased serum creatinine level (mean difference, 0.11 mg/dL; 95% CI 0.07–0.16, $p<0.001$, $I^2=94\%$) when compared with steroid-containing regimens. There is uncertainty of benefit or harm for other outcomes of interest, including incidence of chronic rejection, cytomegalovirus infection, HCV recurrence, malignancy, post-transplant lymphoproliferative disorder, and renal insufficiency.^[107]

Question 10. For maintenance immunosuppression in adult liver transplant recipients, is tacrolimus monotherapy (standard dose or single dose preparation) compared with a dual immunosuppressive drug regimen, as effective in preventing graft loss?

Guidelines statements

29. In the absence of contraindications, tacrolimus monotherapy should be considered as maintenance immunosuppression in patients at low risk for rejection (eg, non-immune-mediated liver disease), as it has similar graft survival when compared with combination therapy. (Strong, Level 3)
30. Combination therapy with tacrolimus and an antimetabolite or a mammalian target of rapamycin inhibitor should be considered for adult liver transplant recipients who require reduced exposure to tacrolimus due to adverse effects (eg, renal dysfunction). (Strong, Level 3)
31. Once-daily extended-release formulations of tacrolimus should be used similarly to twice-daily immediate-release formulations, as they are equally effective for preventing rejection and maintaining allograft survival (Strong, Level 3)

TAC monotherapy is feasible after 3 months post-LT in patients at low risk for rejection, based on a study comparing TAC monotherapy with TAC/glucocorticoid combination therapy without mycophenolate mofetil (MMF).^[108] Nevertheless, TAC is associated with long-term complications, including renal dysfunction, increased risk of malignancy and DM. To reduce TAC exposure, MMF is frequently combined with TAC. Compared with standard-dose TAC, the combination of MMF and reduced-dose TAC improves renal function without increasing graft rejection.^[109] However, MMF is associated with its own set of adverse events that can limit its use, such as gastrointestinal disturbance, leukopenia, thrombocytopenia, anemia, and teratogenicity (Table 4).

The safety of TAC monotherapy ($N=991$) versus TAC/MMF combination therapy ($N=991$) in the first 12 months after LT was assessed in a large study using the Korean national registry database and time-conditional propensity scores. Patients were matched for underlying conditions and time-varying factors, such as liver and renal function, and prior infection at corresponding time points. BPAR (2.6% in TAC vs. 3.5% in TAC/MMF, $p=0.381$) and graft failure rates (0.7% vs. 0.2%, $p=0.08$) were similar in both groups. However,

TABLE 4 Immunosuppressive agents, dosing, common side effects, and drug interactions^[110–117]

Immunosuppressive class	Drugs	Dosing ^a	Trough levels ^a	Common side effects	Drug interactions
Corticosteroids	Solumedrol	500 mg–1 g IV bolus, with progressive taper	—	Diabetes, gastrointestinal, hypertension, obesity, osteopenia, and opportunistic infections	Increased levels with nirmatrelvir/ritonavir
	Prednisone	PO, progressive taper	—		
IL-2 inhibitor	Basiliximab	20 mg IV given at POD 0 and 4	—	Hypersensitivity and opportunistic infections	
Calcineurin inhibitor	Cyclosporine (CSA)	2 mg/kg/d PO twice daily, adjusted to trough levels	250–300 ng/mL for first 3 mo; 100–250 ng/mL month 3 onward	Nephrotoxicity, electrolyte disorders, neurotoxicity, diabetes (more with TAC), hypertension, anemia, gingival hyperplasia (CSA), and hirsutism (CSA)	Increased drug levels with grapefruit juice or co-administration with CYP3A inhibitors: macrolide antibiotics, azole antifungal agents, calcium channel blockers, direct antiviral therapy (with CSA - sofosbuvir, glecaprevir, and voxilaprevir), protease inhibitors, antimalarial agents, benzodiazepines (diazepam and alprazolam), serotonin uptake inhibitors, allopurinol, amiodarone, danazol, proton pump inhibitors, nirmatrelvir–ritonavir. Decreased drug level with CYP3A4 inducers: antimycobacterial agents, anticonvulsants, and the St. John's wort
	Tacrolimus (TAC)	0.1–0.15 mg/kg/d PO twice daily	7–10 ng/mL for first month; 4–8 ng/mL for months 2–12; 4 ng/mL 12 mo onward		
Antimetabolites	Azathioprine	3–5 mg/kg/d PO daily	—	Cytopenias, gastrointestinal, hepatic, hypersensitivity, infections, and malignancy	Increased levels of aminosaliclates may increase cytopenias with angiotensin-converting enzyme inhibitors and ribavirin; may inhibit the anticoagulant effect of warfarin
	Mycophenolate mofetil/ mycophenolic acid (off-label use)	1000–3000 mg PO/360–720 mg PO in divided doses, administered twice daily	—	Cytopenias, gastrointestinal infections, and teratogenicity	Decreased systemic exposure with co-administration of antacids or proton pump inhibitors, drugs that interfere with enterohepatic recirculation, or calcium-free phosphate binders
mTOR inhibitors	Everolimus	1 mg PO bid after month 1	3–8 ng/mL	Gastrointestinal, leukopenia, hyperlipidemia, proteinuria, peripheral edema, wound healing complications, aphthous stomatitis, interstitial pneumonitis, and HAT (sirolimus)	Increased levels with grapefruit juice or co-administration with CSA or CYP3A4 inhibitors: macrolide antibiotics, azole antifungal agents, calcium channel blockers, protease inhibitors, antimalarial agents, benzodiazepines (diazepam and alprazolam), serotonin uptake inhibitors, allopurinol, amiodarone, danazol, and proton pump inhibitors. Decreased levels with CYP3A4 inducers: anticonvulsants and the St. John's wort
	Sirolimus (off-label use)	2 mg PO daily after month 1	4–10 ng/mL		

^aDosing and target trough levels vary at the discretion of the transplant professional.

Abbreviations: CSA, cyclosporine; CYP3A, cytochrome P450 3A; CYP3A4, cytochrome P450 3A4; IV, intravenous; mTOR, mammalian target of rapamycin; PO, oral; POD, postoperative day; TAC, tacrolimus.

the intergroup difference in estimated GFR decline was greater at 6 months (3.1 mL/min/1.73 m², 95% CI 0.8–5.3, $p < 0.01$) in the TAC group compared with the TAC/MMF group.^[118]

Extended-release TAC has also been studied in both the de novo and conversion settings. A meta-analysis of 4 studies ($n = 758$) showed similar BPAR rates in both immediate-release TAC and extended-release TAC (RR 1.06, 95% CI 0.84–1.34, $I^2 = 0\%$) and a similar safety profile.^[119] Moreover, a retrospective analysis of the European Liver Transplant Registry (ELTR) registry ($n = 13,088$) demonstrated better graft survival in LT recipients who were converted from immediate-release TAC to extended-release TAC after 1 month ($p < 0.001$) or were started de novo on extended-release TAC ($p = 0.02$).^[120]

Question 11. *In adult liver transplant recipients with hepatocellular carcinoma as an indication for the LT, does the initiation of a mammalian target of rapamycin inhibitor prevent malignancy recurrence and improve overall patient survival?*

Guidelines statements

32. The addition of or conversion to a mammalian target of rapamycin inhibitor-based immunosuppression may be considered in adult liver transplant recipients transplanted for hepatocellular carcinoma within the Milan criteria to improve recurrence-free survival up to 3 years post-liver transplant. (Weak, Level 3)

Previous studies suggest that the mammalian target of rapamycin (mTOR) pathway is involved in tumor progression and resistance to treatment through the overactivation of several mechanisms.^[121] As a result, targeting the mTOR pathway has emerged as a promising therapeutic strategy in patients with HCC post-LT. Nephrotoxicity is a well-established adverse effect of the long-term utilization of CNI and is the main indication for discontinuing CNI. In this setting, mTOR inhibitors are also an attractive alternative, given their reno-protective effect. Currently, there are 2 oral formulations available: once daily sirolimus (SRL) and twice daily everolimus (EVL). mTOR inhibitors have their own adverse effects, including bone marrow suppression, hypertriglyceridemia, delayed wound healing, pulmonary toxicity, proteinuria, dermatologic toxicity, and edema. In addition, SRL has a black box warning for hepatic artery thrombosis,^[122] but a meta-analysis showed no increased risk with use of EVL^[123] (Table 4).

Clinical studies have shown that SRL-based immunosuppressive regimens decrease the recurrence of post-LT HCC. A meta-analysis of 17 observational and 6 RCTs analyzed the effect of early-initiated (<6 mo post-transplant) mTOR inhibitor-based immunosuppression (mostly used with low-dose CNI) on HCC recurrence and/

or survival in patients transplanted with HCC, with a control arm of CNI-based therapy.^[124] Overall, there were fewer recurrences in the mTOR inhibitor-treated group (RR 0.67, 95% CI 0.56–0.82), but the significant benefit was seen in the 3-year recurrence-free survival (RFS) of cases within the Milan criteria (RR 1.13, 95% CI 1.03–1.23), without benefit in those outside the Milan criteria (RR 0.95, 95% CI 0.83–1.1). Subgroup analysis based on type of mTOR inhibitor showed RFS and overall survival to be slightly better with SRL-based therapy at 1, 3, and 5 years compared with CNI-based therapy, with no significant increase in acute rejection (Table 5). Data for EVL are less compelling, with 1-year and 3-year RFS and overall survival comparable to CNI-treated patients. Overall survival was better with mTOR inhibitor therapy at 1 year (RR 1.07, 95% CI 1.02–1.12), 3 years (RR 1.1, 95% CI 1.02–1.19), and 5 years (RR 1.18, 95% CI 1.08–1.29), although there was significant heterogeneity among pooled studies.

A more recent but smaller meta-analysis (14 cohort studies and 3 RCTs) similarly showed improved RFS with mTOR inhibitor use up to 3 years, as well as better overall survival with mTOR inhibitor use up to 5 years.^[125] On subgroup analysis, SRL similarly had more significant impact on RFS and overall survival than EVL, but interestingly, mTOR inhibitors did not have significant effect on the 3-year overall survival for either within or outside the Milan criteria group, although there was a trend for the within Milan group (RR 1.11, 95% CI 1.0–1.24, $p = 0.06$). This meta-analysis also reported improved renal function with mTOR inhibitor use; there were more adverse effects reported, but no increased HAT incidence in the mTOR inhibitor group.

In a pooled analysis from 2 RCTs ($n = 772$) using EVL/reduced TAC combination versus TAC alone in both DDLT and LDLT patients who had HCC outside Milan criteria, recurrence was not statistically different between the EVL/reduced TAC and the TAC alone groups (5.9% vs. 23.1%, $p = 0.215$), although the recurrence rate in those outside Milan criteria was comparable to that in those within Milan criteria.^[126]

Question 12. *In adult liver transplant recipients experiencing or at risk for calcineurin inhibitor nephrotoxicity, does the initiation of mammalian target of rapamycin inhibitors mitigate the renal dysfunction?*

Guideline statement

33. In adult liver transplant recipients with chronic kidney dysfunction, conversion from calcineurin inhibitors to or addition of mammalian target of rapamycin inhibitor-based immunosuppression should be considered to prevent progression of kidney disease and is best implemented in the early (~3 mo) post-transplant period. (Strong, Level 2)

TABLE 5 Pooled risk ratios of outcomes with mammalian target of rapamycin inhibitors in adult recipients undergoing liver transplant for hepatocellular carcinoma

By mTOR inhibitor	Sirolimus (SRL)		Everolimus (EVL)	
	Studies (n)	Pooled RR (95% CI) of SRL vs. CNI	Studies (n)	Pooled RR (95% CI) of EVL vs. CNI
1-y RFS	9 (1167)	1.12 (1.02–1.22)	2 (228)	1 (0.89–1.13)
3-y RFS	8 (1114)	1.12 (0.99–1.26)	2 (228)	1.06 (0.91–1.25)
5-y RFS	5 (968)	1.15 (0.99–1.35)	0	NA
1-y OS	8 (5204)	1.09 (1.03–1.16)	4 (246)	1.03 (0.95–1.1)
3-y OS	7 (5076)	1.1 (1–1.21)	4 (246)	1.12 (0.95–1.3)
5-y OS	7 (5131)	1.16 (1.07–1.26)	1 (52)	NA
Recurrence rate	9 (5009)	0.69 (0.53–0.88)	9 (803)	0.63 (0.44–0.91)
Overall mortality	5 (4933)	0.63 (0.44–0.92)	4 (512)	0.83 (0.48–1.44)
Recurrence-related mortality	3 (900)	0.47 (0.23–0.96)	3 (159)	0.48 (0.19–1.2)
Acute rejection	5 (984)	1.11 (0.95–1.31)	4 (565)	0.96 (0.57–1.62)
Hepatic artery thrombosis	3 (849)	0.9 (0.25–3.34)	0	NA
By Milan criteria	Within Milan criteria (< 30% subjects outside)		Outside Milan criteria (> 70% subjects outside)	
	Studies (n)	Pooled RR (95% CI) of mTOR vs. CNI ^a	Studies (n)	Pooled RR (95% CI) of mTOR vs. CNI ^a
1-year RFS	3 (556)	1.05 (0.98–1.11)	6 (459)	1.06 (0.94–1.19)
3-year RFS	3 (556)	1.13 (1.03–1.23)	4 (364)	0.95 (0.83–1.1)
5-year RFS	1 (292)	NA	1 (226)	NA
1-year OS	5 (4713)	1.06 (1–1.12)	4 (377)	1.06 (0.94–1.2)
3-year OS	5 (4713)	1.12 (1.01–1.24)	3 (304)	1.06 (0.89–1.28)
5-year OS	3 (4455)	1.15 (1.03–1.28)	2 (268)	1.36 (0.84–2.2)
Recurrence rate	5 (4426)	0.66 (0.44–0.99)	7 (250)	0.77 (0.52–1.13)

Note: Reprinted with permission from Grigg et al.^[124]

^aComparisons between the “within Milan criteria” and “outside Milan criteria” were not statistically significant.

Abbreviations: CNI, calcineurin inhibitor; mTOR, mammalian target of rapamycin; OS, overall survival; RFS, recurrence-free survival; RR, risk ratio.

A significant improvement in renal function was found when SRL was administered to patients with previous HCC who developed nephrotoxicity from CNI, mainly in those who started SRL within 3 months of LT. A meta-analysis of 3 RCTs^[127–129] found significant GFR improvement (10.38 mL/min, 95% CI 3.98–16.77, $P=0\%$) following CNI replacement or reduction with SRL addition between a median of 36–50 months post-LT, although this benefit was non-significant when 8 observational studies were added to the analysis due to study heterogeneity and non-standardized reporting.^[130] A 5-year follow-up of the SILVER study reported improved renal function at 3 months in the SRL arm as compared with the CNI control arm, but the effect was lost thereafter in an intent-to-treat analysis; only those who had durable SRL treatment during the first year after LT had long-lasting preservation of renal function up to 5 years.^[131]

Between the 2 mTOR inhibitors, EVL has more evidence for its renal benefits in LT recipients.^[126,132,133] A meta-analysis of 4 studies ($n=893$) showed significant GFR improvement (by 7.4 mL/min, 95% CI 2.75–17.8) in the EVL arm compared with the control arm, but this effect was greater (by 10.2 mL/min, 95% CI 2.75–17.8, $P=62\%$) when EVL was introduced within 4 weeks after LT. Delayed EVL introduction when GFR is < 60 mL/min per 1.73 m² resulted in only minor improvement in renal function.^[133] Pooled analysis from 2 RCTs ($n=772$) using EVL/reduced-dose TAC versus TAC alone in both DDLT and LDLT showed a lower mean GFR loss (–8.37 vs. –13.40 mL/min/1.73 m², $p=0.001$) with the combination, up to 2 years post-LT.^[126]

Question 13. In adult liver transplant recipients, should protocol liver biopsies or transient elastography be used to monitor long-term liver allograft health?

Guideline statements

34. While liver biopsy has been the gold standard for the diagnosis of post-liver transplant graft dysfunction and may be used before minimization of immunosuppression, protocol liver biopsy is not recommended in the routine management of stable adult liver transplant recipients. (Strong, Level 4)
35. Due to insufficient data, routine use of transient elastography to monitor allograft health cannot be recommended in adult liver transplant recipients. (Weak, Level 3)

Protocol liver biopsy

Few LT centers currently perform protocol liver biopsies in patients with normal liver chemistries, although liver biopsies are safe and can detect subclinical graft injuries that can lead to progressive graft injury.^[134,135] Outside of their potential utility for assessment of progression of disease in the allograft or when reduction of immunosuppression is contemplated, the direct impact of protocol liver biopsies on long-term graft survival remains unproven. A multicenter study of recipients with normal liver tests more than 3 years out from LT ($n=190$) detected portal inflammation, fibrosis, and lobular inflammation in 144 (75.8%), 119 (62.6%), and 115 (60.5%) liver biopsies, respectively.^[136] In another study of LT patients ($n=211$), 32% had minimal histological inflammation, 57% showed significant histological inflammation, and 23% advanced fibrosis ($> F2$) on surveillance biopsies, which were not predicted by lab parameters. Immunosuppression was modified in 79% of patients, and there were no biopsy-related complications.^[135] In another retrospective series of 213 protocol liver biopsies in 110 recipients, there were 9 late-onset acute rejections, 3 isolated perivenular inflammation, 1 plasma cell-rich rejection (PCR), and 1 early chronic ductopenic rejection (CDR) requiring adjustment in immunosuppressive therapy, leading to improvement in the modified hepatic activity index (HAI) score grading in 10/14 cases (71.4%).^[137] Despite these data, the role of protocol liver biopsy in the post-HCV era is not well-defined, and emerging biomarkers and noninvasive tools may provide a more attractive means for monitoring and maintaining allograft health.

Transient elastography

Use of transient elastography in the setting of LT is less well-studied than in non-transplant patients with chronic

liver disease. A meta-analysis of 12 studies evaluating noninvasive methods of assessing biopsy-proven stages 2–4 graft fibrosis post-LT demonstrated the best performance for transient elastography (OR 21.17, 95% CI 14.10–31.77) compared with AST-to-platelet ratio index (APRI) (OR 9.02, 95% CI 5.79–14.07, $p < 0.05$) or fibrosis score-4 (Fib-4) (OR 7.08, 95% CI 4–12.55, $p < 0.05$). However, the heterogeneity of studies led to a wide range of reported threshold values that complicates its use in this population.^[138] Similar findings were reported by a recent single-center retrospective study.^[139] In 2 cohort studies, an elevated liver stiffness has been independently linked to mortality in LT recipients,^[140,141] and single-center studies have shown the utility of transient elastography to detect graft steatosis in LT recipients.^[142,143] However, the definitive role of transient elastography in the management of LT recipients requires further study.

Question 14. Which adult liver transplant recipients can be safely withdrawn from all immunosuppression?

Guideline statement

36. Complete immunosuppression withdrawal is not routinely recommended, given the lack of reliable predictors of withdrawal success and limited data in a very select group of long-term adult recipients. (Strong, Level 4)

Operational tolerance is the survival and functioning of the allograft in the absence of immunosuppression, occurring usually for more than a year, either spontaneously due to non-adherence or on medical advice due to a serious malignancy or infection. In contrast to other solid organ transplants, the liver is more immune-tolerant as it clears blood-borne pathogens and processes gut-derived food-borne antigens. Thus, the chance for operational tolerance is higher in LT than in the other solid organ transplants.^[144] Prospective clinical trials reported that $> 40\%$ of highly selected LT recipients can tolerate complete withdrawal of immunosuppression when performed at a mean of 10.2 years after LT in adults.^[145] However, early immunosuppression withdrawal (within 1–2 years post-LT) is only possible in a very small group of LT recipients.^[146] A recent meta-analysis of 15 studies ($n=620$) showed that the likelihood of operational tolerance increased by 7% (95% CI 4%–10%) per year after LT.^[147]

Long-term studies on the safety of immunosuppression withdrawal regimens are inconclusive, mostly lacking histologic data. Most trials have excluded patients having viral hepatitis or autoimmune liver disease and were performed in the pediatric population rather than in adults. Recent literature has described a variety of clinical, molecular, cellular, and histological

markers originating from the peripheral blood and liver allograft that may help predict who can successfully be weaned off immunosuppression after LT.^[148] However, most of these noninvasive biomarkers have not been prospectively validated. Vionett et al^[136] showed that the severity of clinically silent liver allograft damage was positively associated with TCMR-related transcripts, class II donor-specific antibodies (DSA), ALT, AST, and liver stiffness measurement, and negatively correlated with serum creatinine and tacrolimus trough levels, suggesting that ALT, class II DSA, and liver stiffness measurement may be tools that can identify possible candidates for immunosuppression withdrawal. Specific, reliable biomarkers to identify the ideal candidate for immunosuppression withdrawal, however, are still lacking. Performance of a liver biopsy before consideration of withdrawal appears prudent, as previous immunosuppression withdrawal studies excluded recipients with inflammation, fibrosis, or piecemeal necrosis on histology or with elevated DSA.^[136,146] Furthermore, patients with autoimmune liver disease or a history of rejection are less likely to be withdrawn safely from immunosuppression. Thus, complete immunosuppression withdrawal should not be undertaken except in a closely monitored, highly selected group of LT recipients.

GRAFT REJECTION

Question 15. In adult liver transplant recipients, should plasma cell-rich rejection and late T-cell-mediated rejection be managed differently from acute T-cell-mediated rejection?

Guideline statements

37. In adult liver transplant recipients, the initial approach for mild early and mild late T-cell-mediated rejection should consist of augmentation of existing immunosuppression, including a change from cyclosporine to tacrolimus if applicable. (Strong, Level 3)
38. In addition to augmentation of existing immunosuppression, which includes a change from cyclosporine to tacrolimus if applicable, moderate or severe early or late T-cell-mediated rejection in adult liver transplant recipients should be treated with high-dose intravenous corticosteroids and consideration of addition of another class of immunosuppression if applicable (eg, an antimetabolite or mammalian target of rapamycin inhibitor). (Strong, Level 3)

39. A second course of corticosteroids or lymphocyte-depleting therapy should be considered for adult liver transplant recipients with refractory T-cell-mediated rejection who do not respond to an initial course of corticosteroids. (Strong, Level 4)
40. In adult liver transplant recipients, plasma cell-rich rejection should be treated with corticosteroids and existing immunosuppression should be augmented, followed by consideration of an antimetabolite and/or low-dose maintenance corticosteroid (eg, prednisone 5–10 mg daily). (Strong, Level 4)

Acute TCMR occurs in 10%–30% of liver transplant recipients, usually within the first 3 months of LT.^[111] Late TCMR is defined as occurring 6 months after LT, although others have defined it to occur 3–12 months after LT.^[149,150] Their histologic features are described in [Table 6](#) and depicted in [Figure 2](#). Initial treatment of mild acute TCMR (rejection activity index <4) or late TCMR includes augmentation of existing immunosuppression,^[149,152] although patients with mild acute TCMR rejection may resolve without additional immunosuppression.^[153] Patients on cyclosporine (CSA) should be switched to TAC^[154–157] if there are no contraindications. An additional class of immunosuppression, such as an antimetabolite or mTOR inhibitor, may be added to a CNI-based regimen. Initial treatment of moderate or severe acute or late TCMR includes high-dose corticosteroids (eg, methylprednisolone 500–1000 mg IV daily for 1–3 d),^[158] with a subsequent taper regimen that varies among transplant centers (eg, prednisone 40–80 mg daily, decreased by 10–20 mg decrements until tapered off). Infection prophylaxis should be administered concurrently with high-dose corticosteroids. Approximately 60%–90% of recipients with acute TCMR respond to high-dose corticosteroids,^[158,159] while response rates may be lower for late TCMR. Late TCMR has been associated with lower graft survival rates and the development of ductopenic rejection in those with poor response.^[160] In cases with suboptimal response, repeat liver biopsy may be necessary to exclude other developing etiologies of graft dysfunction, and close collaboration between clinician and pathologist is essential. In recipients who do not respond to the initial course of corticosteroids, a second round of corticosteroids^[158] or administration of lymphocyte-depleting therapy^[161–164] can be considered ([Table 6](#) and [Figure 3](#)). Patient adherence to immunosuppression should be reinforced.

PCR, previously known as de novo autoimmune hepatitis (AIH), is reported in 3%–5% of recipients transplanted for indications other than AIH, with a similar histopathological presentation as “classic” AIH in native livers, except that lymphocytic cholangitis and IgG4-positive rich plasma cell infiltrates are significantly

TABLE 6 Banff criteria for types of rejection^[151] and proposed treatment strategies

Type of rejection	Histologic features on liver biopsy	Proposed treatment
Acute T-cell-mediated rejection (TCMR)	1. Diffuse lymphohistiocytic or polymorphous infiltrate in the portal or perivenular areas 2. Bile duct inflammation and damage, absence of or rare small bile duct loss 3. Subendothelial inflammation or endotheliitis of veins 4. Fibrosis is uncommon	Mild (RAI < 4): • Augment existing immunosuppression, including change from cyclosporine to tacrolimus if applicable • Consider adding a second class of immunosuppressant (antimetabolite or mTOR inhibitor) Moderate or severe (RAI ≥ 4): • High-dose methylprednisolone (500–1000 mg IV) followed by steroid taper • Augment existing immunosuppression, including change from cyclosporine to tacrolimus if applicable • Consider a second class of immunosuppressant (antimetabolite or mTOR inhibitor)
Late TCMR (> 6 mo after liver transplant)	Compared with acute TCMR: 1. Fewer blastic lymphocytes, greater interface and perivenular necroinflammation 2. Less venous endotheliitis 3. Slightly more lobular activity	Same as acute TCMR
Plasma cell-rich rejection	1. Prominent plasma cells (> 30%) in the portal or perivenular infiltrates, with easily recognizable periportal/interface and/or perivenular necroinflammatory activity, usually involving a majority of portal tracts and/or central veins 2. Enrichment by IgG4-positive plasma cells in some cases 3. Prevalent central perivenulitis 4. Higher incidence of portal microvascular endothelial cell C4d deposition Other findings: • Presence of donor-specific antibodies • Indication for liver transplant was not autoimmune hepatitis	• Corticosteroids • Augment existing immunosuppression • Consider: ◦ Maintaining on low-dose prednisone ◦ Adding an antimetabolite
Chronic ductopenic rejection (CDR)	1. Bile duct atrophy/pyknosis affecting the majority of bile ducts, with or without bile duct loss 2. Foam cell obliterative arteriopathy or bile duct loss in > 50% of the portal tracts 3. Perivenular fibrosis Factors associated with progression: • Centrilobular necroinflammation • Hepatocyte necrosis and apoptosis • Loss of hepatic artery branches from portal tracts	• Prevention and effective treatment of acute TCMR may reduce risk of CDR • Augment existing immunosuppression, including change from cyclosporine to tacrolimus • Consider adding a second class of immunosuppressant (antimetabolite or mTOR inhibitor) • Steroids or anti-thymocyte globulin are not effective • Recipients with < 50% bile duct loss may have more reversibility.
Acute antibody-mediated rejection	1. Endothelial hypertrophy 2. Portal capillary dilatation 3. Microvasculitis 4. Periportal edema 5. C4d staining in sinusoidal endothelium Other findings: • Elevated donor-specific antibody titers • Unexplained graft dysfunction • Thrombocytopenia • Hypocomplementemia	• High-dose glucocorticoids • Augment existing immunosuppression, including change from cyclosporine to tacrolimus if applicable • Consider a second class of immunosuppressant (antimetabolite or mTOR inhibitor) Refractory cases: • IV immunoglobulin and plasma exchange with or without anti-CD20 agents
Chronic antibody-mediated rejection	1. Unexplained mononuclear and/or perivenular inflammation 2. Low-grade interface and/or perivenular necroinflammatory activity 3. Moderate portal/periportal sinusoidal and/or perivenular fibrosis 4. Focal C4d staining in > 10% of portal tract endothelia 5. Nodular regenerative hyperplasia and obliterative arteriopathy in some cases Other findings: • Elevated donor-specific antibody titers in some cases	• Augment existing immunosuppression, including a change from cyclosporine to tacrolimus if applicable • Consider a second class of immunosuppressant (antimetabolite or mTOR inhibitor)

Abbreviations: C4d, complement component 4d; CDR, chronic ductopenic rejection; CNI, calcineurin inhibitor; DSA, donor-specific antibody; IV, intravenous; MFI, mean fluorescence intensity; mTOR, mammalian target of rapamycin; RAI, rejection activity index; TCMR, T-cell-mediated rejection.

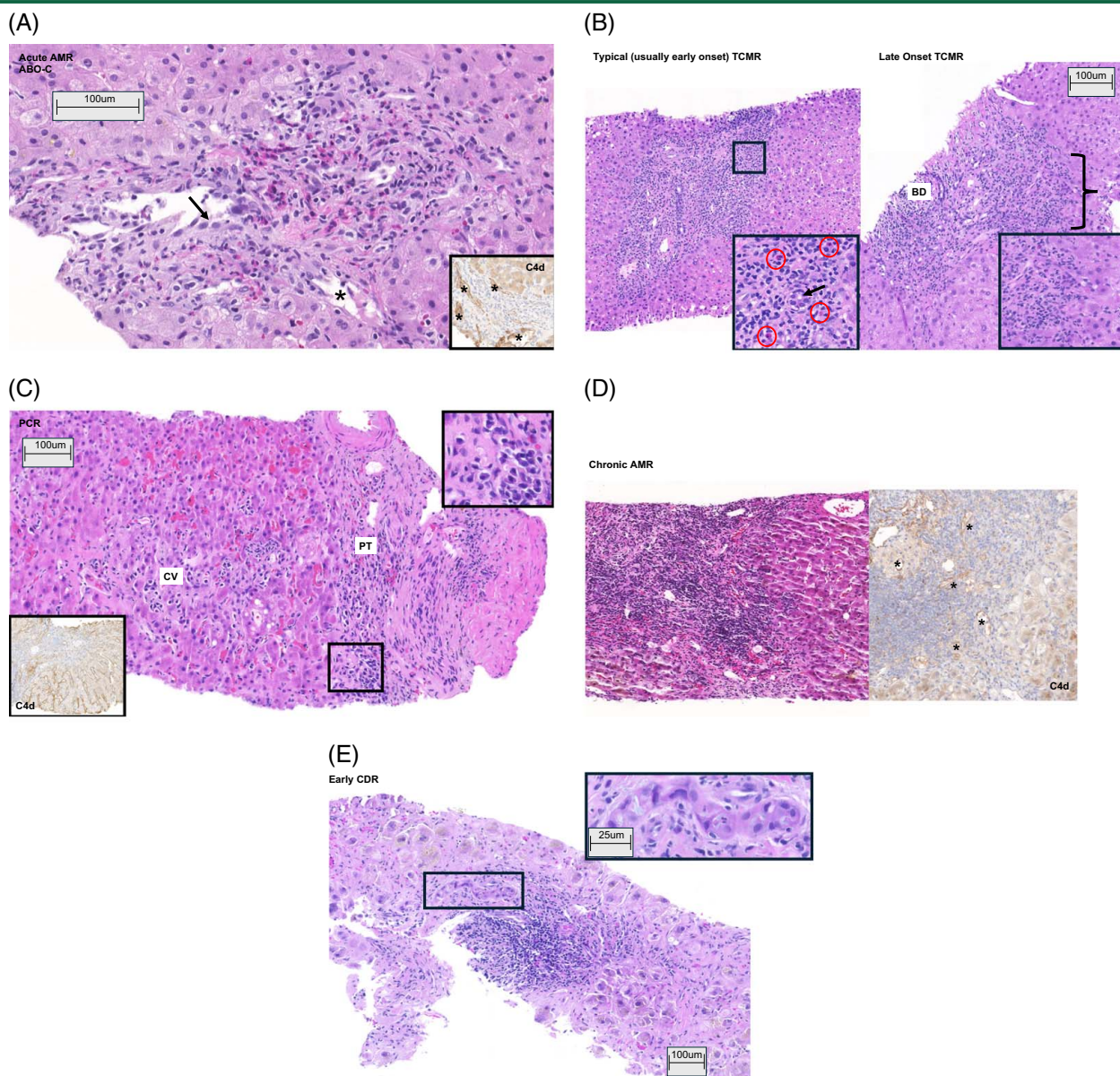
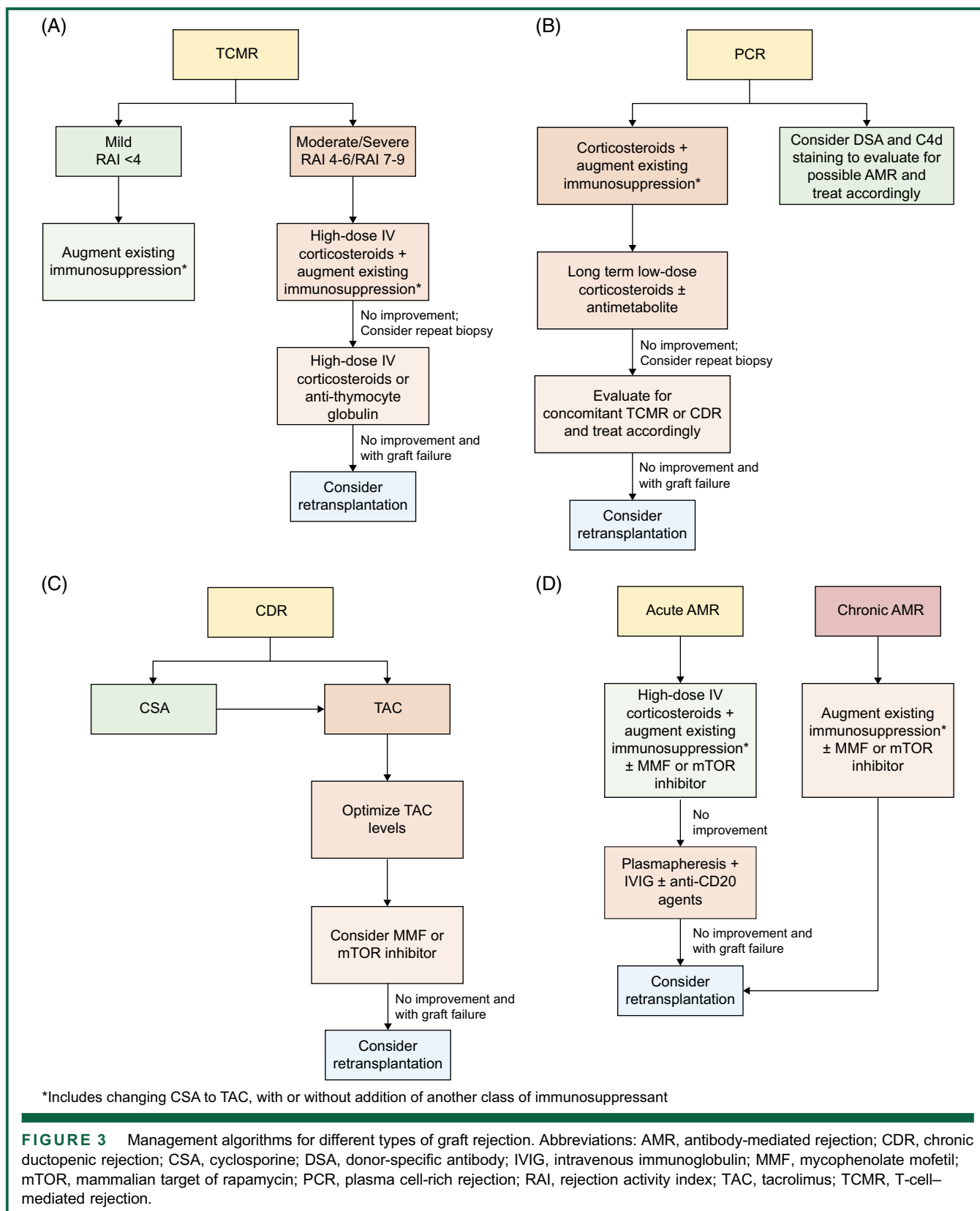


FIGURE 2 Types of liver allograft rejection. (A) Acute AMR. Representative portal tract from liver allograft biopsy showing acute AMR in a highly sensitized, ABO-compatible recipient with the DSA showing multiple class I and II specificities, most of which were > 10,000 MFI. Note the marked endothelial cell hypertrophy in the portal vein branch (arrow) and inlet venules (*) with co-existent microvasculitis. The portal inflammatory infiltrate is focally enriched with eosinophils, and C4d staining (lower right inset) shows diffuse staining of the portal microvascular endothelium (portal vein, portal capillaries, and inlet venules). (B) Typical T-cell-mediated rejection (TCMR). Comparison of typical TCMR that occurs mostly early after transplantation with "late onset" TCMR that usually occurs 6 months or more after transplantation. The major differences include more pleomorphic portal infiltrates, including eosinophils (red circles); lymphocytic cholangitis (arrow in inset); and subendothelial infiltration of portal and central vein branches in typical TCMR, whereas late onset usually shows a paucity of lymphocytic cholangitis among the bile ducts (BD), less pleomorphic infiltrate with more severe and prevalent interface (bracket and inset) and perivenular necroinflammatory activity. (C) PCR. Biopsy with PCR showing plasma cell-rich (> 30%) portal and perivenular inflammation with interface and perivenular necroinflammatory activity. The upper right inset illustrates the area captured by the box (lower right) at higher magnification to illustrate the plasma cell-rich nature of the inflammation. The lower left inset shows diffuse portal and sinusoidal microvascular endothelial cell C4d positivity in this strongly DSA-positive recipient. (D) Chronic AMR. Typical changes of chronic AMR combined with late onset TCMR. Note the PT inflammation with interface necroinflammatory activity (left frame; bracket), portal microvascular inflammation (right frame; *), and diffuse portal capillary and inlet venule C4d positivity (right frame; *). (E) Early CDR. The hallmark histopathological finding in early chronic ductopenic rejection is biliary epithelial cell senescence-related changes (box and upper right inset) characterized by cytoplasmic eosinophilia, irregular nuclear spacing, and occasional biliary epithelial cell apoptosis. Abbreviations: AMR, antibody-mediated rejection; CDR, chronic ductopenic rejection; CV, central vein; PCR, plasma cell-rich rejection; PT, portal tract.



more prevalent in PCR.^[151,159,165] Compared with acute TCMR, PCR is characterized by (1) significantly greater plasma cell infiltrate, (2) more common enrichment by IgG4-positive plasma cells, (3) more prevalent central

perivenulitis, (4) increased incidence of DSA, and (5) higher incidence of portal microvascular endothelial cell complement 4d (C4d) deposition^[110,151,166] (Figure 2). In 1 study, PCR occurred later than TCMR at a median of

56 months versus 15 months, and PCR had a higher rate of advanced fibrosis at 47% versus 2%, respectively.^[150] It is not clear if PCR is an atypical presentation of acute or chronic AMR or a separate entity by itself.^[166] In a single-center study (n=35), 65.7% of patients had at least 1 prior episode of TCMR before being diagnosed with PCR; DSAs were positive in 6/19 patients, and C4d immunostaining was positive in 9/10 patients.^[167] Thus, evaluation for DSA and C4d staining should be considered to detect a possible co-existent AMR component.^[149,152,168]

Studies on the optimal management of PCR are lacking but augmentation of immunosuppression in addition to corticosteroids are commonly the first line of treatment for PCR, with consideration of the addition of an antimetabolite.^[154,169,170] In 1 small case series, maintenance doses of prednisone and/or antimetabolites were associated with a lower risk of relapse and a decreased risk of progressive liver disease, re-LT, and mortality.^[154] Data are lacking on the efficacy of SRL and EVL for PCR. As with other cases of rejection, adherence to immunosuppression should be reinforced.

Question 16. Are peripheral blood biomarkers, compared with liver biopsies, reliable indicators of T-cell-mediated rejection and other forms of allograft injury in adult liver transplant recipients?

Guideline statements

41. In adult liver transplant recipients, peripheral blood biomarkers alone should not be used to diagnose or exclude T-cell-mediated rejection and other forms of allograft injury. (Strong, Level 4)
42. In adult liver transplant recipients, peripheral blood biomarkers that measure intracellular ATP after CD4 cell activation may be considered to determine the immunological status of the transplant recipient, risk of infection, and need for adjustment of immunosuppression. (Weak, Level 3)

While liver chemistry tests allow for routine monitoring of the liver allograft, there are no well-established normal ranges for them in the post-LT setting.^[171] Advancements in molecular biology technology have been pivotal in the development of promising peripheral blood biomarkers to detect TCMR and other forms of allograft injury as a noninvasive alternative to liver biopsy. Types of peripheral blood biomarkers are summarized in [Table 7](#) and have been reviewed elsewhere,^[148,191] but most biomarkers are not approved for LT recipients and have not been validated.

DSA is a type of peripheral blood biomarker and is discussed under questions 18 and 19.

Mixed lymphocyte reaction assays assess immunologic function in LT recipients, but are labor-intensive, time-consuming, and not reliably reproducible. A meta-analysis of a Food and Drug Administration-approved in vitro assay (ImmuKnow, ViraCor-IBT Laboratories) that measures intracellular ATP after CD4 cell activation reported a sensitivity, specificity, and summary receiver operator curve of 65.6%, 80.4%, and 0.835, respectively, for detecting acute rejection when compared with liver biopsy, although there was significant heterogeneity among studies.^[172] A trial that randomized LT recipients to standard practice or serial immune function assay compared rates of infection, rejection, and patient survival between the groups, in which the TAC dose was adjusted based on the assay result in the monitored group. The monitored group had lower rates of infection (42% vs. 54.9%, $p < 0.05$), higher 1-year patient survival (95% vs. 82%, $p < 0.01$), and similar rates of rejection as the control group (19.0% vs. 13.7%, $p = \text{NS}$).^[192] However, these studies were performed during the pre-DAA era, and the results may not be as applicable in the current LT population.

Peripheral blood gene expression profiling and donor-derived cell-free DNA have been studied in the characterization of allograft rejection and immunosuppression optimization. These noninvasive biomarkers are utilized in the management of heart and kidney transplant recipients and have been explored in LT, but small sample sizes and the lack of proper control groups and independent validations have limited their adoption in routine clinical use. Studies evaluating the transcriptomic profiles associated with biological fluids have revealed the potential for non-coding RNA to serve as biomarkers of graft injury. Of these, circulating microRNAs have been most extensively studied as markers of TCMR.^[193,194] Donor-derived cell-free DNA is another promising biomarker for the detection of graft injury. Two small studies have demonstrated the utility of serial plasma donor-derived cell-free DNA levels in detecting preclinical TCMR before any rise in the liver chemistry tests.^[190,195] These tools hold promise but need to be validated in larger patient populations.

Question 17. In adult liver transplant recipients, can chronic ductopenic rejection be reversed with immunosuppressive therapy?

Guideline statements

43. In adult liver transplant recipients, chronic ductopenic rejection should be treated early (when there is < 50% bile duct loss or ductopenia on liver biopsy) with augmentation of existing immunosuppression, in order to maximize treatment response and reversibility. (Strong, Level 4)

TABLE 7 Peripheral blood biomarkers under investigation as indicators for allograft rejection in adult liver transplant recipients

Assay	Source	Study population*	Results
Intracellular ATP in activated CD4+ cells ^[172]	Whole blood	• Meta-analysis of 5 studies on performance for rejection • 4 studies used biopsy-proven acute rejection (BPAR)	• AUC = 0.835 for ACR
Preformed DSA ^[173]	Serum	• N = 110	• Preformed class I and II HLA-DSA with MFI > 1000 had the best positive predictive value for ACR
Preformed DSA ^[174]	Serum	• N = 177	• Preformed DSA with MFI > 1500 was not associated with ACR
De novo DSA ^[175]	Serum	• N = 27 with de novo DSA • N = 53 without de novo DSA	• No association between de novo DSA and ACR. • Recipients with ACR had a higher total number of antibody-verified class I MM eplets
SNPs in CD161 for killer cell lectin-like receptor B1 polymorphism ^[176]	Not specified	• N = 178 with ACR • N = 163 without rejection	• No difference in frequency of killer cell lectin-like receptor polymorphism between the 2 groups
CD16+ monocytes ^[177]	Whole blood flow cytometry	• N = 25 with ACR • N = 20 recipients no ACR	• Frequency of CD16+ monocytes is higher in the acute TCMR group vs. the no ACR group (45% and 7.17%, $p < 0.001$)
Circulating immune complexes ^[178]	Serum	• N = 10 LDLT with acute ACR • N = 22 LDLT without ACR	• Thrombospondin-1, apolipoprotein-E, apolipoprotein C-II and complement factor-H were only detected in the acute TCMR group
miRNAs ^[179]	Plasma	• N = 17 with ACR • N = 8 with subclinical rejection • N = 120 with no rejection	• Significantly higher levels of specific miRNAs pre-LT and post-LT in acute ACR and subclinical rejection groups
SNPs in genes that regulate TH1, TH2, and Tregs: TBC21, GATA3, and FOXP3 ^[180]	Not specified	• N = 49 with late ACR (> 31 d after LT) • N = 149 without late ACR	• Higher frequency of FOX3 SNP in the late ACR group than in without late ACR group
SNPs in chemokine receptors 9 and 10 ^[181]	Whole blood DNA	• N = 59 with ACR • N = 156 without ACR	• No association between CXCL9/10 polymorphisms and ACR • CXCL9 genotype AA developed ACR earlier
SNPs in circulating T-follicular cells ^[182]	Plasma	• N = 12 with ACR • N = 25 without ACR	• SNPs in T-follicular cells associated with ACR
Lymphocyte and monocyte phenotypes and NK cells ^[183]	Whole blood flow cytometry	• N = 72	• No association with ACR and CD4+, CD8+, CD45+, CD64+, monocyte HLA-DR expression, NK cells
T-cell subsets Interleukin-6 and 17 ^[184]	Whole blood and serum flow cytometry	• N = 67 with early ACR (within 1 y) • N = 28 with normal liver tests	• The following markers did not predict ACR: CD4, CD8, CD4+/CD25+/FOXP3+, IFN-gamma, IL-6, IL-17
SNPs in CD28 ^[185]	Whole blood DNA	• N = 144 with ACR • N = 147 without ACR	• SNP in CD28 was not associated with ACR
SNPs in C3 allotypes ^[186]	Whole blood DNA	• N = 30 LDLT	• C3-SF and C3-FF genotypes associated with ACR (6.25-fold higher than C3-SS allele)
36 gene panel ^[187]	Blood gene expression via microarray RNA profiling	• N = 60 from 2 cohorts with ACR • N = 66 with elevated liver tests from other causes • N = 95 with normal liver tests	• Accuracy was 0.77 for ACR in the validation cohort • Genes upregulated in ACR mapped to hepatic proliferation and inflammatory pathways
Liver-derived extracellular particles ^[188]	Plasma	• N = 12 with acute ACR • N = 24 without ACR • N = 19 non-transplant patients as control	• ASGR1+CD130+AnnV+ extracellular particles have the highest accuracy for ACR with AUC 0.80, sensitivity 100%, specificity 68.5%
De novo DSA ^[189]	Blood	• N = 32	• No association between de novo DSA and graft injury

TABLE 7. (continued)

Assay	Source	Study population*	Results
Donor-derived cell-free DNA ^[190]	Plasma	• N = 57 with ACR • N = 94 with normal graft function • N = 68 with acute dysfunction but no rejection	• ACR vs. no ACR—sensitivity 87.5%, specificity 80%

Abbreviations: ACR, acute cellular rejection; DSA, donor-specific antibody; IFN, interferon; MFI, mean fluorescent intensity; LDLT, living donor liver transplant; LT, liver transplant; miRNA, microRNA; SNP, single-nucleotide polymorphism.

44. Adult liver transplant recipients with chronic ductopenic rejection who are taking cyclosporine should be switched to tacrolimus if there are no contraindications. (Strong, Level 3)
45. Mycophenolate mofetil or a mammalian target of rapamycin inhibitor may be added in adult liver transplant recipients with chronic ductopenic rejection who are on tacrolimus monotherapy. (Weak, Level 4)
46. Adult liver transplant recipients with progressive chronic ductopenic rejection despite adjustments in immunosuppressive therapy should be considered for retransplantation. (Strong, Level 3)

CDR can result from severe or recurrent episodes of acute rejection with potentially irreversible damage to bile ducts, arteries, and veins.^[196] Other risk factors for CDR include young recipient age, male-to-female sex mismatch, a history of AIH or biliary disease, non-Caucasian race, donor age older than 40 years, and presence of DSA.^[197–200] Small bile duct damage and loss, and perivenular fibrosis are essential for the diagnosis (Table 6 and Figure 2). Compared with TCMR, CDR has less inflammation, more fibrosis, and is less reversible. Early CDR is characterized by <50% bile duct loss, and late CDR is characterized by ≥50% bile duct loss and progression of perivenular fibrosis.^[199]

Initial treatment for CDR includes augmentation of existing immunosuppression and/or adding immunosuppressants. In a multicenter cohort study, 70% of patients with CDR responded to conversion of CSA to TAC,^[201] supporting this as a viable management strategy.^[159] For those on TAC monotherapy, MMF^[202] or an mTOR inhibitor^[203–205] can be added to the regimen. Steroids or anti-thymocyte globulin has not been shown to be effective. Adherence to immunosuppression should be reinforced. Early CDR may be reversible, with response more likely to be seen in those with bile duct loss in <50% of portal tracts, mild perivenular fibrosis, and portal tract hepatic arteriole loss in <25% of portal tracts.^[199] Progressive CDR that

fails to respond to modification of immunosuppression and leads to allograft failure may require re-LT.^[201]

Question 18. In adult liver transplant recipients, how are acute and chronic antibody-mediated rejection diagnosed and managed?

Guideline statements

47. In adult liver transplant recipients, acute antibody-mediated rejection should be suspected in a recipient with unexplained graft dysfunction, thrombocytopenia, and hypocomplementemia. Donor-specific antibody testing and liver biopsy should be obtained with C4d staining to assess for signature histologic features. (Strong, Level 3)
48. Initial treatment of acute antibody-mediated rejection in adult liver transplant recipients should include high-dose corticosteroids, augmentation of existing immunosuppression with or without the addition of an immunosuppressant from a different class. (Strong, Level 3)
49. In adult liver transplant recipients, treatment with intravenous immunoglobulin and plasma exchange with or without anti-CD20 agents may be considered for acute antibody-mediated rejection refractory to initial therapy. (Weak, Level 4)
50. In adult liver transplant recipients, the diagnosis of chronic antibody-mediated rejection is challenging; workup should include donor-specific antibodies and typical findings on liver biopsy. Other liver diseases that may cause a similar pattern of liver injury should be excluded. (Strong, Level 4)
51. In adult liver transplant recipients, augmentation of existing immunosuppression, with or without the addition of a second class of immunosuppressant, should be considered as initial therapy for chronic antibody-mediated rejection. (Strong, Level 5)

52. There are insufficient data to recommend the use of anti-CD20 therapy, proteasome therapy, plasma exchange, or intravenous immunoglobulin in adult liver transplant recipients with chronic antibody-mediated rejection. (Strong, Level 4)
53. In adult liver transplant recipients, retransplantation should be considered for acute or chronic antibody-mediated rejection that results in graft failure. (Strong, Level 5)

Acute Antibody-mediated Rejection

Acute AMR may cause unexplained graft dysfunction, thrombocytopenia, hypocomplementemia, and vasculitis on liver biopsy with diffuse C4d staining.^[206] Acute AMR can occur in ABO-incompatible grafts but is rare in ABO-compatible LT recipients, because of the regenerative capacity of the liver, large sinusoidal network, and secretion of soluble class I human leukocyte antigen (HLA) that binds DSA.^[151,166] In addition, variable hepatic microvasculature class II expression provides fewer class II DSA targets. Clinically, AMR should be considered in recipients with graft dysfunction in the setting of high sensitization who develop otherwise unexplained thrombocytopenia, low-complement levels, and circulating immune complexes. The diagnosis of acute AMR can be made by signature histologic features (Table 6 and Figure 2) in the presence of high DSA. Staining of portal vein and capillaries and sinusoidal endothelial cells for C4d is most specific for acute AMR.^[151,206]

Most studies on treatment for acute AMR in LT are retrospective and limited to small single-center series, pediatric recipients, or in ABO-incompatible LT. Initial management includes high-dose corticosteroids and augmenting existing immunosuppression, with or without adding an immunosuppressant from a different class. Plasma exchange and intravenous immunoglobulin, with or without rituximab, bortezomib, and/or eculizumab, have been used for recipients who do not respond to initial therapy (Table 8), but data are lacking on their efficacy and safety.^[166] Re-LT may be considered for acute AMR that results in graft failure.

Chronic Antibody-mediated Rejection

Chronic AMR occurs in <10% of LT recipients and typically is associated with de novo DSA to HLA class II, especially DQ.^[151,217] Typical histologic features for chronic AMR are difficult to characterize primarily because most episodes represent mixed chronic AMR and TCMR (Table 6 and Figure 2). The chronic AMR score was developed to identify recipients at high risk for graft loss, consisting of histologic findings, the presence of DSA, and elimination of other potential

etiologies that cause a similar pattern of liver injury. The chronic AMR score has been associated with graft loss with an AUC ranging from 0.70 to 0.81.^[218]

Initial treatment for chronic AMR includes increasing plasma tacrolimus levels with or without the addition of a second class of immunosuppressant.^[166] The role of anti-CD20 agents, proteasome inhibitors, or plasma exchange and intravenous immunoglobulin therapy is uncertain, due to the lack of data in LT recipients with chronic AMR.^[210,215,216] Re-LT should be considered for chronic AMR that results in graft failure.

The optimal immunosuppression strategy for recipients who require re-LT, particularly for those with graft loss from immunologic causes, is not established. The only study on immunosuppression following re-LT was a retrospective analysis of the SRTR database (n=3483), which reported recurrent disease (51.3%) and rejection (15.2%) requiring re-LT performed >365 days after the primary LT. These re-LT recipients were more likely to receive induction therapy and were more frequently discharged on triple therapy with CNI, antimetabolites, and steroids compared with primary LT recipients ($p < 0.001$). There was considerable heterogeneity in the maintenance regimens according to the initial graft failure cause, with no clear trends, but steroid-based regimens with or without antimetabolites were maintained throughout the first year for patients having autoimmune liver disease. Although there was a trend linking lymphocyte-depleting induction therapy and better patient survival for the entire re-LT group, this was not observed in the later re-LT group.^[219] In cases of recurrent graft failure requiring multiple re-LT, the role of immune dysregulation causing unrecognized AMR needs to be considered, which in turn, merits careful consideration of the choice and intensity of immunosuppression.^[220]

Question 19. In adult liver transplant recipients, what is the significance of donor-specific antibodies, and how should their surveillance be conducted?

Guideline statements

54. In adult liver transplant recipients, donor-specific antibodies should be assessed in the setting of severe or steroid-resistant rejection, plasma cell-rich rejection, antibody-mediated rejection, or allograft dysfunction of unknown etiology. (Strong, Level 3)
55. Since high donor-specific antibodies are not always associated with graft complications, and mean fluorescence intensity thresholds and measurement techniques have not been standardized, routine pre-transplant donor-specific antibody testing for all adult liver transplant candidates cannot be recommended. (Weak, Level 3)

TABLE 8 Summary of case series of treatment for acute antibody-mediated rejection after liver transplantation in adults

Study	n	Treatment-n	Outcomes-n (time from transplant)
Kamar et al (2009) ^[207]	2	Corticosteroids, OKT3, PP, rituximab-1 Corticosteroids, PP, rituximab-1	Dead-1 (POD 41) Alive-1 (9 mo)
Kozlowski et al (2011) ^[208]	3	IVIG, PP, rituximab-1 ATG, PP, IVIG, rituximab-1 IVIG, OKT3, rituximab, PP-1	Dead-2 (40 and 116 d) Retransplantation-1 (13 mo)
Chan et al (2011) ^[209]	5	Corticosteroids-3 Corticosteroids, Rituximab, PP-1 Corticosteroids, PP, bortezomib-1	Dead-4 (11–18 d) Alive-1 (180 d)
Paterno et al (2012) ^[210]	3	Bortezomib-2 Rituximab, Bortezomib, PP-1	Alive-3 (2.5 mo, 1 y, 2.5 y)
Lee (2012) ^[211]	9	Corticosteroids, rituximab, PP-3 Corticosteroids, rituximab, PP, bortezomib-6	Dead-6 (11–63 d) Alive-3 (18–26 mo)
Del Bello et al (2017) ^[212]	9	Corticosteroids, PP, IVIG-3 Corticosteroids, PP, IVIG, ATG-2 Corticosteroids, PP, ATG-1 Corticosteroids, rituximab-2 Corticosteroids, augmentation of immunosuppression-1	Dead-2 (5–9 mo after AMR) Re-LT and alive-1 (7 mo after AMR) Alive-6 (4–90 mo after AMR)
Vandendorpe et al (2018) ^[213]	4	Corticosteroids, rituximab, PP, IVIG	Dead-1 (13 mo) Alive-3 (30–70 mo)
Del Bello et al (2020) ^[214]	10	ATG-3 Corticosteroids, rituximab, PP-3 Corticosteroids, augmentation of immunosuppression-2 Unspecified-2	Dead-2 Alive-8 (1–125 mo after AMR)
Sakamoto et al (2021) ^[215]	3	Corticosteroids, rituximab, PP- 1 Corticosteroids, rituximab, PP, IVIG-2	Dead-2 (12–191 d) Alive-1 (3384 d)
Dumortier et al (2023) ^[216]	19	Most received corticosteroids, rituximab, PP, IVIG	Dead-7 Re-LT and alive-1 Alive-11

Abbreviations: ATG, anti-thymocyte globulin; AMR, antibody-mediated rejection; IVIG, intravenous immunoglobulin; LT, liver transplantation; POD, post-operative day; PP, plasmapheresis.

56. In adult liver transplant recipients, the performance of serial measurement of donor-specific antibodies can be individualized in the setting of verified or suspected antibody-mediated rejection that is being treated or monitored. (Weak, Level 4)

DSA are recipient antibodies generated against HLA classes 1 and 2 of the donor liver or other donor antigens.^[221] DSA may be preformed or develop de novo, with multi-specificity and higher mean fluorescence intensity (MFI) being associated with greater rates of rejection and reduced graft survival.^[166] Preformed DSA with potential clinical significance has been tentatively defined as MFI ≥ 5000 at the 2015 Banff Conference on Allograft Pathology,^[151] but positivity cutoff varies by laboratory and in different studies, so the optimal range is still to be determined.

Most liver recipients do not have preformed DSA, unless they have had prior alloantigen exposure through pregnancy, transfusion, or prior transplantation.^[166] Low-level preformed

mostly class I DSA may be absorbed post-LT, but the persistence of preformed DSA has been associated with higher rates of rejection at 1 year, with increased risks for graft loss and CDR.^[173,214,222] However, many patients with high DSA may not develop these complications,^[174] so routine assessment of DSA in every recipient cannot be justified.

The incidence of de novo DSAs in LT recipients is lower than that for other solid organ recipients, being 12.2% at 1 year, 13.4% at 5 years, and 19.5% at 10 years. These are usually directed toward class II HLA antigens,^[223] which show lower expression in structurally normal allografts compared with inflamed allografts.^[224] Graft pathology increases class II expression/DSA targets, and therefore, the possibility of further target tissue damage.^[218] Risk factors for de novo DSA are similar to pre-sensitization and include female sex, longer time between transplant and DSA screening, lower total tacrolimus exposure,^[213] and purposeful lowering of immunosuppression. A recent meta-analysis of 15 studies reported the presence of de novo DSA with a cutoff of 3000 MFI to be associated with

increased risk of graft rejection (OR 6.43, 95% CI 3.17–13.04, $p < 0.001$, $I^2 = 50\%$) as well as graft loss and CDR (OR 3.61, 95% CI 1.94–6.71, $p < 0.001$, $I^2 = 58\%$) when compared with recipients without de novo DSA. The associations between de novo DSA and overall poorer outcomes were consistent across all subgroups and sensitivity analyses.^[225]

Suboptimal immunosuppression and non-adherence increase the risk of developing de novo DSA.^[213,225] A prospective study comparing 45 matched DSA-positive recipients to 45 DSA-negative recipients demonstrated subtle differences in inflammation and unique patterns of fibrosis despite normal or near normal liver tests, and led to the development of the chronic AMR score.^[218] There is no agreed-upon MFI cutoff value that correlates with chronic AMR, although most studies have shown correlations with AMR at MFI > 5000 .^[221] The presence of de novo or preformed C1q+ and C3d +-binding antibodies to HLA class II antigens has also been associated with AMR.^[226] Regardless, the presence of HLA and non-HLA DSA in long-term surviving LT recipients portends increased liver allograft inflammation and fibrosis as well as decreased survival.^[227] Future studies should examine the association between DSA and decreased allograft survival, especially since some patients can have high DSA yet still have a functioning, healthy allograft. Despite the scarcity of data to dictate the frequency, serial DSA measurements to guide immunosuppression management in cases of suspected or diagnosed AMR are reasonable.

RECURRENT DISEASE

Viral hepatitis

Hepatitis B

Question 20. In adult liver transplant recipients without hepatitis B infection who received hepatitis B surface antigen-negative, anti-hepatitis B core-positive grafts, what is the optimal approach to prevent hepatitis B infection post-transplantation?

Guideline statement

57. In adult liver transplant recipients who are hepatitis B surface antigen-negative and who receive anti-hepatitis B core-positive grafts, first-line prophylaxis with either entecavir, tenofovir disoproxil, or tenofovir alafenamide should be administered indefinitely to prevent reactivation. (Strong, Level 3)

HBsAg-negative recipients who receive a HBsAg-negative, HBcAb-positive graft have a risk of transmission up to 75%, with higher risks in recipients who are

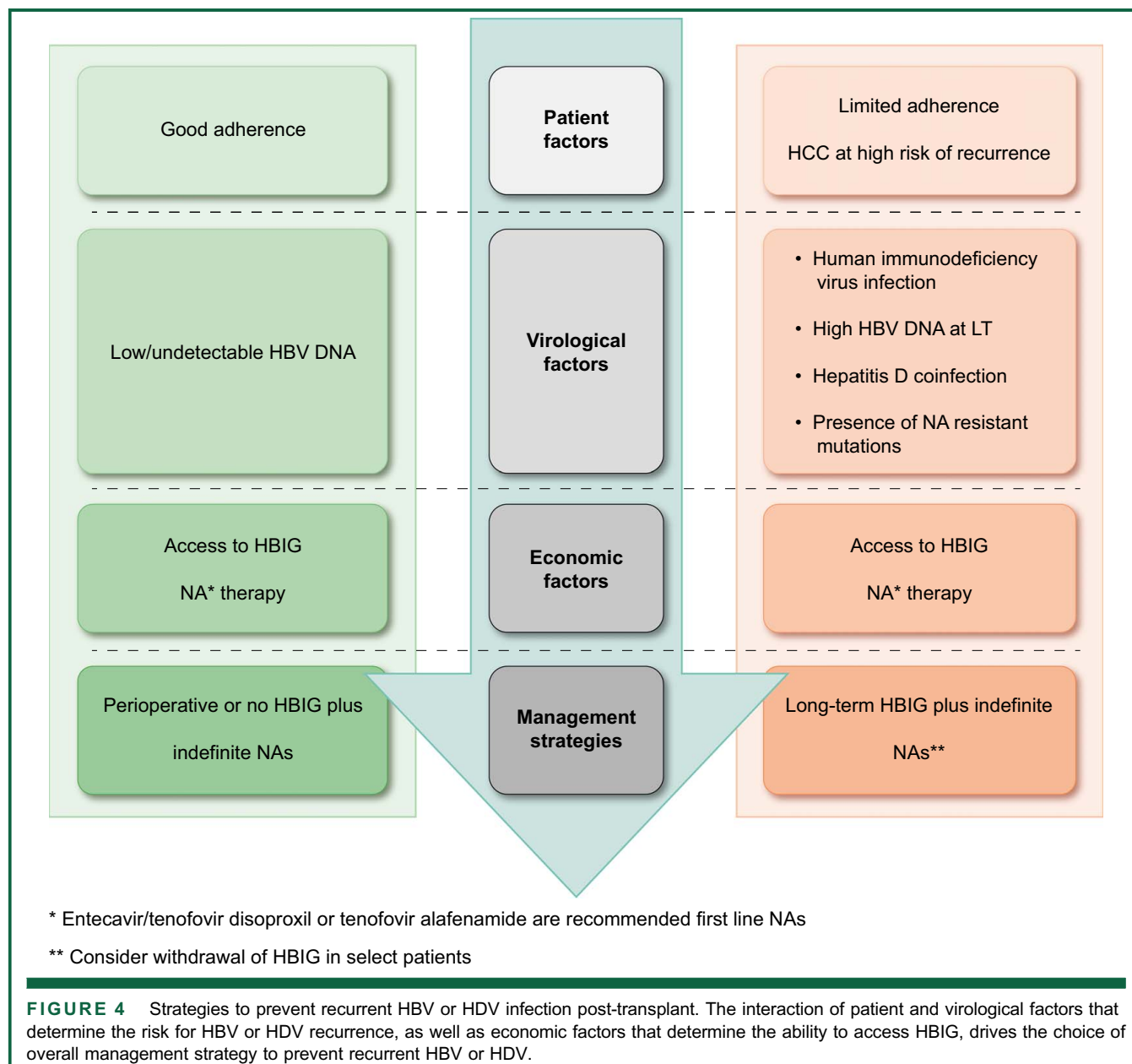
HBsAb and HBcAb negative and lower in those who are HBsAb-positive.^[228] Antiviral therapy is highly effective in preventing infection and should be administered immediately post-LT and continued indefinitely. There is no requirement for hepatitis B immunoglobulin (HBIG) in this setting.^[229] Although a few HBsAg-negative recipients who received HBcAb-positive grafts ($n = 6$) were able to safely discontinue antiviral therapy in the setting of HBsAb titers > 36.6 IU/mL, this strategy needs to be validated in a larger population.^[230] Those who develop de novo HBV infection should remain on first-line nucleos(t)ide analogue (NA) therapy lifelong.

Question 21. In adult liver transplant recipients with chronic hepatitis B infection with or without hepatitis D co-infection, what is the optimal post-transplant management strategy to prevent recurrent hepatitis B or recurrent hepatitis B/hepatitis D co-infection?

Guideline statements

58. In all adult liver transplant recipients with hepatitis B infection, first-line nucleos(t)ide analogue should be administered to prevent recurrent hepatitis B infection post-liver transplant. (Strong, Level 3)
59. In adult liver transplant recipients with hepatitis B infection alone, the addition of hepatitis B immunoglobulin to first-line nucleos(t)ide analogue therapy post-liver transplant should be individualized, with low-risk recipients receiving hepatitis B immunoglobulin for short-term or no hepatitis B immunoglobulin, and high-risk recipients receiving longer-term hepatitis B immunoglobulin. (Strong, Level 2)
60. In all adult liver transplant recipients with hepatitis B and hepatitis D co-infection, hepatitis B immunoglobulin should be given in addition to first-line nucleos(t)ide analogue. (Strong, Level 4)

First-line nucleos(t)ide analogue (NA) therapy should be initiated before LT in candidates with chronic HBV infection and is also the treatment of choice for HBV recurrence or de novo infection in LT recipients.^[228] The use of HBIG in addition to NA to prevent the recurrence of HBV infection varies from center to center. A recent meta-analysis (25 studies, almost all observational, $n = 2327$) reported similar HBV recurrence rates between patients who received NA plus a finite course of HBIG as compared with those with indefinite combination therapy ($p = 0.48$), but HBV recurrence was significantly higher in patients who received NA monotherapy ($p < 0.001$). Notably, the 2 studies that administered NA monotherapy had higher pre-LT HBV DNA levels, which themselves may have increased the recurrence risk. In addition, the presence of HCC before LT



was also identified as a risk factor for HBV recurrence in this meta-analysis.^[231] In contrast, a larger meta-analysis (41 studies, n=7897) reported comparable 10-year HBV recurrence rates and 7-year survival between recipients who received HBIG plus NA as compared with those who received NA monotherapy.^[232] Other studies supporting HBIG withdrawal and continuation of NA have also shown comparable results with indefinite HBIG plus NA, depending on the risk of recurrent HBV infection, but the duration of HBIG administration varied widely between 6 months to several years.^[233,234] A preliminary report noted that complete withdrawal of HBIG and NA therapy is feasible and safe in low-risk HBsAg-positive recipients and is associated with HBsAb seroconversion;^[235] future studies are needed to define specific management strategies. The use of long-term HBIG (intravenous, subcutaneous, or intramuscular) is individualized depending on risks of reinfection based on

HBV DNA level, co-infection with HDV or HIV, center preference, and other factors^[229,236] (Figure 4).

A single-center report of 179 individuals with HBV infection who underwent LT and received 5000 units HBIG in the anhepatic phase and 400 IU intramuscularly every 2 weeks in combination with NA, reported HBV recurrence in 6 individuals, 5 of whom underwent LT due to HCC and the other patient had HBV/hepatitis delta virus (HDV) co-infection.^[237] There are no approved therapies for HDV post-LT, and treatment relies on control and prevention of HBV. In a study comparing the outcomes of 127 individuals with HBV infection to 128 individuals with HBV/HDV infection, all of whom were treated with HBIG and tenofovir, there was no difference in HBsAg recurrence, and there were no cases of HDV recurrence.^[238] Patients with HDV who undergo transplant should receive HBIG plus NA therapy. There are

limited reports suggesting withdrawal of HBIG is feasible with low rates of HDV recurrence and no difference in graft or patient outcomes compared with HBV/HDV subjects who continue HBIG.^[235,239–241]

Question 22. In adult liver transplant recipients without hepatitis B infection, can hepatitis B surface antigen-positive grafts, along with antiviral therapy, be used in hepatitis B-negative recipients?

Guideline statement

61. In adult liver transplant recipients without hepatitis B infection, hepatitis B surface antigen-positive grafts may be considered for use with first-line nucleos(t)ide analogue therapy to suppress viremia, with or without hepatitis B immunoglobulin, after a thorough discussion on risks and benefits between the candidate and the transplant team. (Weak, Level 4)

Although up to 2 million individuals in the United States may be chronically infected with HBV, there are limited observational data on the use of HBsAg-positive grafts in HBsAg-negative recipients. A retrospective cohort study of the OPTN database from 1999 to 2021 examined outcomes in 265 recipients of HBV-positive grafts, defined as HBsAg and/or HBV nucleic acid test (NAT)-positive, of whom 198 were HBV-negative at the time of transplant. In a matched analysis of 209 recipients with known HBV status, there was no difference in recipient mortality (log-rank $p=0.47$) or graft loss (log-rank $p=0.72$) after a mean follow-up of 3.3 years, with 1-year and 3-year survival rates of 93.1% and 84.8%. Similar results were seen in a subgroup analysis of 119 HBV NAT-positive grafts with no differences in graft or patient survival compared with those who received HBV NAT-negative grafts. Only 1 patient had viral hepatitis listed as the cause of death in 2006, which was at the time of introduction of the first NA with a higher barrier to resistance, entecavir.^[242]

A national registry cohort from China reported comparable patient ($p=0.6$) and graft ($p=0.65$) survival at up to 60 months in 55 HBsAg-negative recipients undergoing DDLT with HBsAg-positive grafts, as compared with those who received HBsAg-negative grafts.^[243] Another case series examined outcomes in 33 HBsAg-negative recipients of HBV NAT-positive grafts, comparing HBsAb-positive recipients on entecavir monotherapy, and HBsAb-negative recipients on entecavir and HBIG. No differences in patient or graft survival were noted by Kaplan–Meier survival analysis. Detectable HBV DNA was observed in 9/33 subjects post-transplant and at 1 year, 8/33 had positive HBsAg, with 23 having evidence of immunity or resolved infection.^[244]

All recipients of HBV-positive grafts, defined as HBsAg and/or HBV NAT positive, should receive first-

line NA therapy with high barrier to resistance including entecavir, tenofovir alafenamide (off-label use) or tenofovir disoproxil, with or without HBIG, to suppress viremia. The optimal use of HBIG remains to be determined in this population, but it may be given in those without HBsAb at the time of LT at the physician's discretion. A recent observational study compared outcomes of patients undergoing LT who received entecavir or tenofovir disoproxil. No differences in patient or graft outcomes were noted, although tenofovir disoproxil therapy was associated with a higher risk of renal dysfunction, defined as a 20% persistent reduction in eGFR compared with baseline at a median of 34 months of follow-up. Careful attention to monitoring renal function should be undertaken in those receiving tenofovir disoproxil.^[245] Ongoing HCC screening should be considered for these recipients until further data become available.

Hepatitis C

Question 23. In adult liver transplant recipients with hepatitis C viremia prior to transplant who receive hepatitis C-viremic grafts, what is the optimal treatment strategy to achieve sustained virologic response post-transplant?

Guideline statements

62. In adult liver transplant recipients with chronic hepatitis C infection who receive hepatitis C-viremic grafts, direct-acting antiviral treatment should be initiated post-transplantation for a duration of 12 weeks without ribavirin, preferably as soon as stabilized from the surgical procedure and within the first 3 months. (Strong, Level 3)
63. In adult liver transplant recipients who have hepatitis C viremia and receive a hepatitis C-viremic graft, because of the potential for change in genotype from the hepatitis C donor, hepatitis C genotyping should be performed if not treated with pan-genotypic direct-acting antiviral therapy. (Strong, Level 3)
64. In adult liver transplant recipients who are being treated with direct-acting antiviral therapy, assessment and monitoring for potential direct-acting antiviral therapy interactions with calcineurin inhibitors and other medications should be performed during and after treatment. (Strong, Level 3)

The safety and efficacy of post-LT HCV therapy with DAA are well-established, and recommended DAA

regimens for treatment-inexperienced and treatment-experienced individuals are outlined in the AASLD and Infectious Diseases Society of America Hepatitis C Guidance.^[246] All HCV-viremic LT recipients should receive DAA to achieve a cure. Assessment for drug–drug interactions between DAA and immunosuppressive agents is essential in the LT population (Table 4), and dosing adjustments may be necessary as informed by the guidance document.^[246]

Question 24. In adult liver transplant recipients without chronic hepatitis C infection who receive anti-hepatitis C-positive and nucleic acid testing-positive grafts, what is the optimal strategy to treat hepatitis C infection post-transplantation?

Guideline statements

65. In adult liver transplant recipients without hepatitis C infection who receive a hepatitis C-viremic graft, pan-genotypic direct-acting antiviral therapy should be initiated as soon as the patient is stabilized from the surgical procedure. (Strong, Level 3)
66. In adult liver transplant recipients without hepatitis C infection who receive an anti-hepatitis C-positive, nucleic acid test-negative graft, transmission of hepatitis C is < 10% and monitoring of hepatitis C RNA testing with a reactive strategy is appropriate. (Strong, Level 4)

Due to the high efficacy of DAA therapies and the persistent donor shortage, combined with an increase in opioid overdose deaths, HCV-positive donors have been increasingly used in LT recipients without HCV infection, which can shorten wait times and expand the donor pool with excellent short-term results.^[247] LT programs should have a protocol that ensures informed consent and a pathway for DAA therapy and appropriate follow-up. Genotyping and viral load quantification are not medically required to initiate DAA treatment with pan-genotypic coverage. Assessment for drug–drug interactions between DAA and immunosuppressive agents (Table 4) is essential, as discussed in the Hepatitis C Guidance 2023 Update.^[246] Multiple large case series have demonstrated the efficacy of DAA therapy post-LT.^[248–251] While an initial study reported a higher rate of AMR, subsequent studies have not shown similar findings.^[251] One prospective multicenter trial treated 13 HCV-negative LT recipients at a mean time of 7 days post-LT, with all subjects achieving sustained response.^[252] In a single-center study of 21 recipients without HCV infection who received anti-HCV-positive but HCV-NAT-negative donors, only 2 (9.5%)

developed HCV viremia post-LT;^[253] thus, in this population, close monitoring for recurrent viremia is appropriate.

Immune-mediated diseases

Question 25. Do recipients with autoimmune hepatitis, primary biliary cholangitis, or primary sclerosing cholangitis, as indications for liver transplant, compared with other liver transplant recipients, require more intense or specific immunosuppressive regimens to prevent disease recurrence and rejection?

Guideline statements

67. Adult liver transplant recipients should not be maintained on long-term corticosteroids to prevent recurrent autoimmune hepatitis. (Strong, Level 3)
68. Given insufficient data, the use of antimetabolites to prevent recurrent disease in adult liver transplant recipients transplanted for autoimmune hepatitis cannot be routinely recommended. (Weak, Level 3)
69. Cyclosporine-based immunosuppression in place of a tacrolimus-based immunosuppression to lower the risk of recurrent primary biliary cholangitis is not recommended, due to concerns for a higher risk of graft rejection. (Strong, Level 3)
70. In adult liver transplant recipients, prophylactic use of ursodeoxycholic acid may be considered to prevent recurrent primary biliary cholangitis. (Weak, Level 3)
71. Due to insufficient data, the choice of immunosuppressive regimen in adult liver transplant recipients transplanted for primary sclerosing cholangitis should be individualized. (Strong, Level 4)

Recurrent AIH, primary biliary cholangitis (PBC), and primary sclerosing cholangitis (PSC) can lead to graft loss and mortality.^[254] The incidence and diagnostic criteria for disease recurrence are outlined in Table 9.

Recurrent Autoimmune Hepatitis

Outcomes of LT for AIH are typically good, although infection is relatively common in the early post-LT period due to immunosuppression usage.^[255–258] Patients transplanted for AIH-related acute liver failure have similar

TABLE 9 Frequency and diagnostic criteria for recurrent autoimmune hepatitis, primary biliary cholangitis, and primary sclerosing cholangitis

Disease	Frequency of recurrence	Diagnostic criteria for recurrence
Recurrent autoimmune hepatitis	17%–50% ^[255–260]	History: Confirmed diagnosis of autoimmune hepatitis prior to LT AND Histology: Plasma cell predominant hepatitis involving portal tracts in the absence of endothelial cell and bile duct damage typical of ACR
Recurrent primary biliary cholangitis	17–53% ^[261–263]	History: Confirmed diagnosis of primary biliary cholangitis prior to LT AND Histology: Granulomatous cholangitis with bile duct injury and portal inflammation with the exclusion of histological features for rejection
Recurrent primary sclerosing cholangitis	10%–37% ^[254]	Modified from Graziadei criteria ^a History: Confirmed diagnosis of primary sclerosing cholangitis prior to LT AND Imaging: Cholangiography demonstrating intrahepatic and/or extrahepatic biliary stricturing, beading, and irregularity > 90 d OR Histology: Fibrous cholangitis and/or fibro-obliterative lesions with or without ductopenia, biliary fibrosis, or biliary cirrhosis In absence of other known causes of biliary strictures: Hepatic artery thrombosis/stenosis Established ductopenic rejection Anastomotic strictures alone Non-anastomotic strictures before post-LT day 90 ABO incompatibility between donor and recipient Donation after circulatory death related cholangiopathy Covid-related cholangiopathy Other causes of ischemic cholangiopathy IgG4 disease

^aAdapted from Graziadei et al.^[264]

Abbreviations: IgG4, immunoglobulin G4; LT, liver transplantation.

survival and recurrence rates as those with AIH-related cirrhosis. However, TCMR and CDR occur more commonly in recipients with AIH than other liver diseases^[257] and allograft failure can result from CDR or cirrhosis due to recurrent AIH.^[257,258] A small, single-center study showed low-dose corticosteroid in combination with other immunosuppressive agents reduced recurrent AIH,^[265] but a meta-analysis of limited datasets and another more recent study demonstrated no benefit of long-term administration of corticosteroids on disease recurrence^[257,266] or patient survival.^[257] Ongoing corticosteroid use may in fact be harmful for patients who have underlying osteoporosis or are at increased risk of infection. In a French database study, initial use of an antimetabolite at LT was protective against graft rejection, but data on its benefit as a maintenance immunosuppressant were lacking.^[256] In a large international multicenter cohort study, MMF use was associated with a higher risk of recurrent AIH, while there was no difference between CSA and TAC.^[258] Like AIH, recurrent AIH is treated with corticosteroids and azathioprine or MMF.^[257]

Recurrent PBC

Recurrent PBC requires histological confirmation (Table 9), so its incidence may be underestimated in

long-term LT survivors who have mild ALP elevations but no liver biopsy. PBC patients are at higher risk of rejection, including late TCMR, which may complicate the diagnosis of recurrent PBC.^[160] Recurrent PBC has not been associated with decreased survival in the past, but a recent multicenter study suggested that graft loss and death do indeed occur.^[261] A meta-analysis showed TAC use was associated with a higher recurrent PBC rate (HR 2.62, 95% CI 1.35–5.09, $p = 0.004$, $I^2 = 54\%$), while use of MMF or azathioprine had no significant effect.^[262] However, a recent OPTN database study reported graft loss from recurrent PBC to be similar between recipients receiving CSA and TAC (TAC HR 0.90, 95% CI 0.49–1.63, $p = 0.72$). TAC avoidance may have potentially increased the risk for rejection, although the difference was not statistically significant between TAC and CSA (23% vs. 30%, $p = 0.48$).^[267] A more recent ELTR database study also reported similar findings, with comparable graft loss (aHR 1.07, 95% CI 0.92–1.25, $p = 0.402$) and death (aHR 1.06, 95% CI 0.90–1.24, $p = 0.473$) between TAC or CSA-treated LT recipients. Interestingly, there was less graft loss (aHR 0.72, 95% CI 0.6–0.87, $p < 0.001$) or death (aHR 0.72, 95% CI 0.59–0.87, $p < 0.001$) with MMF, while there was increased graft loss (aHR 1.31, 95% CI 1.13–1.52, $p < 0.001$) and death (aHR 1.34, 95% CI 1.15–1.56, $p < 0.001$) with corticosteroid use.^[268] In the previous meta-analysis, prophylactic ursodeoxycholic

acid started within the first 2 weeks post-LT was protective (HR 0.40, 95% CI 0.28–0.57, $p < 0.001$, $I^2 = 0\%$).^[262] Ursodeoxycholic acid was also associated with decreased graft loss (aHR 0.33, 95% CI 0.13–0.82, $p < 0.05$) and death (aHR 0.69, 95% CI 0.49–0.96, $p < 0.05$) in another study ($n = 780$).^[269] Similar to PBC, recurrent PBC should be treated with ursodeoxycholic acid.^[270] At this time, the safety and efficacy of the second-line agents, elafibranor or seladelpar, for recurrent PBC are not known.

Recurrent Primary Sclerosing Cholangitis

TCMR risk in PSC patients is higher at up to 70%,^[271] as compared with other indications for LT. However, CSA and TAC did not differ in TCMR risk in a retrospective propensity score-matched, registry-based study. Interestingly, this study showed a lower graft loss or death rate with a TAC plus MMF regimen.^[272] A meta-analysis showed that TCMR and multiple TCMR episodes were associated with recurrent PSC. Colectomy before LT was protective, but the primary immunosuppressive regimen had no significant effect.^[273] An increased risk for recurrent PSC related to multiple episodes of TCMR was also noted in another meta-analysis, without a significant difference between CSA and TAC use.^[259] A more recent international multicenter retrospective study also showed a higher risk of recurrent PSC with multiple TCMR.^[274] In the latter meta-analysis, colectomy before LT was protective, but inflammatory bowel disease presence itself was not a significant predictor for recurrent PSC, although only 4 studies were included in the analysis.^[259] Nonetheless, a dual-center Australian study reported a significant impact of inflammatory bowel disease activity, with moderate to severe disease at any time post-LT carrying a higher risk of recurrent PSC.^[175] The use of biologic agents for inflammatory bowel disease concomitant with immunosuppression has been reported only in case reports and case series.^[276–279] Clinical response can be seen in 65% of cases,^[279] but 26%–47% had significant infections, and 5% developed colorectal cancer.^[276] There remains no specific effective medical therapy, including ursodiol, as prophylaxis for or treatment of recurrent PSC to date.

METABOLIC DISEASES

Recurrent metabolic dysfunction–associated steatohepatitis

Question 26. Do recipients with metabolic dysfunction–associated steatohepatitis as the indication for liver transplantation have similar outcomes compared with

recipients without metabolic dysfunction–associated steatohepatitis?

Guideline statement

72. Liver transplantation should be considered for adult patients with metabolic dysfunction–associated steatohepatitis, as they have similar post-transplant outcomes as patients without metabolic dysfunction–associated steatohepatitis, up to 10 years. (Strong, Level 2)

MASLD confers a higher risk for cardiovascular disease, extrahepatic cancers, type 2 DM, and chronic kidney disease, all of which are also common complications in patients after LT and can lead to increased mortality rates.^[280] In addition, metabolic dysfunction–associated steatohepatitis (MASH) patients tend to be older and have higher body mass index, DM, and PVT at the time of LT, which are all associated with worse outcomes. However, a meta-analysis of non-randomized trials ($n = 22$ trials, 1538 MASH vs. 6014 non-MASH patients) showed no difference in 1-year and 5-year graft and patient survival after LT.^[281] While individual studies using SRTR data have shown variable post-LT outcomes of MASH patients, a meta-analysis of 15 studies ($n = 119,327$, 20.2% with MASH) showed comparable complications, overall survival, and graft survival rates compared with non-MASH patients. The pooled 1-year, 5-year, and 10-year all-cause mortality rate in MASH patients after LT was 12.5%, 24.4%, and 37.9%, respectively, comparable to non-MASH patients (HR 0.910, 95% CI 0.760–1.10, $p = 0.34$).^[282] The leading causes of death for patients transplanted for MASH are listed in Table 10. The prevention and management of recurrent MASLD centers on optimization of risk factors, which is addressed in the upcoming “AASLD Guideline on Adult Liver Transplantation: Diagnosis and Post-Transplant Management of Non-Graft-Related Complications.”

Question 27. Do specific immunosuppressive agents affect the outcomes of adult patients transplanted for metabolic dysfunction–associated steatohepatitis?

Guideline statements

73. The use of corticosteroids beyond 3–6 months should be avoided in adult liver transplant recipients transplanted for metabolic dysfunction–associated steatohepatitis due to a higher risk of post-transplant diabetes and metabolic syndrome. (Strong, Level 3)

TABLE 10 Causes of mortality in patients transplanted for suspected metabolic dysfunction-associated steatohepatitis (MASH)

Cause of mortality	SRTR (n = 850) [283]	ELTR (n = 631) [284]
Infection	135 (15.9%)	114 (18%)
Cardiovascular	149 (17.6%)	32 (5.1%)
Cancer (non-liver)	79 (9.3%)	35 (5.5%)
Cancer (recurrent HCC)	–	59 (9.4%)
Graft failure	53 (6.2%)	61 (9.7%)
Recurrent MASH	–	8
Rejection	–	18
Other causes	53	35
Multiple organ failure	98 (11.5%)	–
Pulmonary embolism	57 (6.7%)	–
GI complications	19 (2.2%)	17 (2.7%)
Perioperative (including PNF)	–	31 (4.9%)
Renal failure	21 (2.5%)	–
Others	88 (10.3%)	16 (2.5%)
Unknown	151 (17.8%)	266 (42.2%)

Abbreviations: ELTR, European Liver Transplant Registry; MASH, metabolic dysfunction-associated with steatohepatitis; PNF, primary nonfunction; SRTR, Scientific Registry for Transplant Recipients.

74. Due to the contribution of calcineurin inhibitors to the risk of de novo diabetes and metabolic syndrome, lowering calcineurin inhibitor exposure may be considered in adult liver recipients transplanted for metabolic dysfunction-associated steatohepatitis, with or without concurrent use of mycophenolate mofetil. (Weak, Level 5)

In a meta-analysis, MASLD recurred in 59%, 57%, and 82% and recurrent MASH was seen in 53%, 57%, and 38% of cases at 1, 3, and 5 years following LT, respectively.^[285] De novo MASLD occurred in 67%, 40%, and 78%, and de novo MASH occurred in 13%, 16%, and 17%. Corticosteroids are associated with DM, hypertension, hyperlipidemia, steatosis, and weight gain. Steroid-free immunosuppressive protocols have shown less de novo DM and hypertension post-LT.^[96,97,99] TAC increases the risk of de novo DM, and CSA increases the risk of hypertension, hyperlipidemia, and metabolic syndrome after LT.^[286–288] mTOR inhibitors are also associated with hyperlipidemia and hyperglycemia, but less weight gain.^[289,290] The only agent not associated with the development of metabolic syndrome is MMF,^[286] making it an attractive adjunct agent to calcineurin inhibitor regimens in LT recipients transplanted for MASH. There are very limited data on immunosuppression and MASLD recurrence, with 2 small studies implicating corticosteroids as a risk factor for the

development of recurrent MASLD,^[291,292] while 1 study did not.^[293] Thus, these limited data suggest that corticosteroids may increase the risk of recurrent MASLD and should not be used long term.

Prevention of recurrent MASLD following LT is closely linked to the prevention or management of metabolic syndrome in the LT recipient. While imaging studies obtained in the post-LT course may suggest the recurrence of MASLD, a liver biopsy remains the gold standard for assessment of MASH and for other graft pathology.^[294] Although AASLD recommends FIB-4 as a risk assessment tool for MASLD in the general population,^[295] this and most other clinical prognostication models utilize platelet counts, which do not reliably reflect liver fibrosis in the LT population. A meta-analysis has demonstrated better performance of transient elastography for graft fibrosis assessment in LT recipients as compared with FIB-4 and APRI score, but most studies focused on recurrent HCV as the underlying disease.^[138] Single-center cohort studies have suggested the utility of transient elastography in detecting graft steatosis in LT recipients,^[142,143] and further studies are needed to define its role, along with other noninvasive measures of liver stiffness, specifically in the LT population.

Recurrent alcohol-associated liver disease

Question 28. For adult recipients undergoing liver transplant for alcohol-associated liver disease, are routine monitoring (clinical interview, toxicology, psychometrics), medications, and other non-medical interventions effective in preventing alcohol relapse?

Guideline statements

75. Regular monitoring for alcohol use in at-risk recipients is recommended, given clinical outcome risks and known rates of recurrent post-transplant alcohol-associated liver disease, which occur with variable onset and trajectories. (Strong, Level 5)
76. Although data support the use of alcohol use disorder medications in patients with alcohol-associated liver disease, they are not well-studied in adult liver transplant recipients, and their impact on clinical outcomes is not known. However, their use can be considered as part of multidisciplinary post-transplant management. (Strong, Level 5)
77. A multidisciplinary approach to patient care, alcohol risk monitoring, and intervention when relapse occurs is recommended in adult liver recipients transplanted for alcohol-associated liver disease. (Strong, Level 4)

Routine monitoring

There is no consensus about acceptable levels of post-LT alcohol use.^[296] Five-year post-LT alcohol relapse rates range between 20% and 50%.^[297] Not all alcohol slips or relapses lead to unfavorable outcomes, yet some relapse patterns (ie, early-onset post-LT drinking) are associated with steatohepatitis and acute rejection on biopsy more often, as well as higher rates of graft failure.^[298] Sustained post-LT alcohol use is a major predictor of post-LT mortality.^[299] Alcohol sobriety tends to become stable and durable at 5 years,^[300] a benchmark many ALD patients will not have achieved at the time of LT or in the early postoperative years. This is particularly true for patients with decompensated alcohol-associated cirrhosis with recent or active alcohol use or those with acute alcohol-associated hepatitis (AAH).

Early LT for patients with short sobriety is increasingly common since a seminal study established its feasibility in highly selected AAH candidates in 2011.^[301] Clinical benefit of early LT was also demonstrated among 147 AAH recipients from the ACCELERATE-AH study with a pre-LT sobriety median of 55 days, in which 1-year post-LT survival was 94% and 3-year survival was 84%; only 11% of patients returned to sustained alcohol use.^[302] A cohort of AAH patients evaluated for early LT in the QuickTrans study had higher alcohol relapse rates among early LT recipients ($n=68$) versus standard sobriety interval recipients ($n=93$, absolute difference 9.1%, 95% CI $-\infty$ to 21.1, $p=0.45$) and higher 2-year rates of high alcohol intake (absolute difference 16.7%, 95% CI 5.8–27.6) despite similar 2-year survival rates.^[303] However, another cohort study that reported similar 1-year allograft (92.7% vs. 90.5%, $p=0.42$) and patient survival (94.1% vs. 95.9%, $p=0.60$) also demonstrated a similar relapse-free (80.4% vs. 83.5%, $p=0.41$) and hazardous relapse-free survival (85.8% vs. 89.6%, $p=0.41$) between LT recipients with short sobriety (<6 mo) and standard sobriety (≥ 6 mo).^[304] These data support previous guidelines, which assert that pre-LT alcohol sobriety length is not a reliable parameter to be used alone during LT candidate selection and that certain carefully selected early LT candidates not responding to medical therapy can become successful recipients.^[305] However, routine post-LT monitoring for recurrent alcohol use remains prudent.

Detecting slips in any patient can help prevent progression to full relapse,^[297] meaning LT teams should strive to detect any alcohol use in at-risk recipients. Alcohol relapse occurs with different frequencies and severities at different times across an LT course.^[298–300,306] Existing literature regarding post-LT relapse tends to examine pre-LT predictors involved in candidate selection [eg, duration of sobriety, level of daily drinking, duration of heavy drinking, persistent drinking through liver disease, psychiatric comorbidity, other substance use, prior relapses and unsuccessful rehabilitation attempts, level of social support, degree of

adherence, family history of alcohol use disorder (AUD), etc.].^[296,298,299,306,307] Fewer studies examine post-LT predictors of post-LT drinking.^[297] Examples of predictive post-LT medical and psychosocial variables include poorer perceived post-LT health when compared with 1-year prior, increased physical pain, more fatigue/less energy, chronic post-LT complications (eg, biliary strictures, incisional hernias), increased stress, or pre-LT other substance use.^[298,300]

Not all of these factors are directly alcohol-associated, which is likely why studies support early recognition and response via multidisciplinary approaches as superior methods by which to address post-LT relapse.^[297,306,308] Patients at high risk for post-LT alcohol use may require close follow-up regardless of any post-LT challenges that arise. However, the effects of post-LT chronic medical complications may be most pronounced in patients deemed low or moderate risk for post-LT alcohol use.^[306] More studies are required to understand how to risk-stratify such patients and optimally respond to their needs.

Some studies show elevated scores pre-LT on scales, including High-Risk Alcohol Relapse Scale (HRAR), Alcohol Use Disorder Identification Test (AUDIT), Sustained Alcohol Use Post-Liver Transplant (SALT), or Rochester Relapse Risk (R_3S), are predictive of post-LT relapse.^[296,298,299,308,309] Their potential use as a prospective monitoring tool post-LT, as opposed to a pre-LT screening instrument, is unclear.

Regular use of serum phosphatidylethanol (PEth), given its high sensitivity and specificity and wide 2–4 week detection window, can reassure a team that they are not missing alcohol relapses.^[300,309] Urine ethyl glucuronide, another alcohol biomarker with a narrower detection window of 3–5 days, can be used instead of or in addition to PEth.^[306] Other patterns of substance use can be associated with post-LT drinking,^[307] making use of other toxicology (eg, nicotine metabolites, urine drug screens) potentially important for detecting and responding to post-LT relapse risk. Detailed, consensus protocols about post-LT toxicology use do not exist, and toxicology is often used at the discretion of LT clinicians.^[306] A proposed testing for recipients at high-risk for post-LT relapse is a PEth assay every 3 months for 2 years.^[306]

Medications

Few studies exist that investigate the effects of embedded addiction care, including medication use, on LT outcomes like post-LT alcohol use.^[300] A pilot study ($n=30$) suggested that acamprosate is safe in LT recipients, but its impact on LT outcomes remains unclear.^[310] More studies are needed to investigate the impact of on-label (eg, naltrexone, acamprosate) and off-label (eg, gabapentin, baclofen, topiramate) AUD medications on post-LT alcohol use.

Non-medical interventions

“Structured management” and “multidisciplinary support” have been shown to reduce post-LT relapse rates.^[297,300] These clinical approaches generally consist of embedded, prospective psychosocial evaluation and care within the LT center throughout all phases of pre-LT and post-LT care, regular clinic visits with psychosocial specialists, and frequent monitoring using toxicology and validated questionnaires. When compared with a historical cohort who did not receive “multidisciplinary support,” alcohol relapse rates were significantly lower than controls (HR 0.21, 95% CI 0.06–0.68, $p < 0.01$) while mortality rates were similar ($p = 0.38$). Another study found no differences in survival rates between short sobriety patients receiving integrated AUD treatment and those who had 6 months or more of sobriety.^[311] Only 6.8% of short sobriety patients relapsed to alcohol post-LT, a rate lower than their standard sobriety interval counterparts (16%).

In summary, regular monitoring for alcohol use in recipients at risk for alcohol relapse allows for early detection of risks and for intervention that may prevent progression to full relapse. This may be achieved through alcohol questionnaires and/or alcohol biomarkers such as serum PETH and urine ethylglucuronide.^[305] Pharmacotherapy for AUD in LT recipients has not been well-studied, but its use can be considered as part of multidisciplinary post-transplant management.

FUTURE DIRECTIONS

The field of LT continues to rapidly evolve. During the time it took to develop these guidelines, several new investigations encompassing the different forms of machine perfusion were published, and exciting new advances in xenotransplantation occurred. The current guidelines were formulated using the best available evidence at the time of writing to guide transplant clinicians in the management of the LT recipient. However, there certainly remain areas where data are not robust or available, and clinical expertise and experience have driven the current standard of practice. This is where the art of medicine meets the science.

Since allograft survival is no longer impaired by HCV recurrence, transplant clinicians can now focus on improving and maintaining allograft health, long-term clinical outcomes, and recipient quality of life. The Banff Working Group on Liver Allograft Pathology meetings will continue to fine-tune the impact of late TCMR on patient outcomes. Peripheral blood biomarkers bring a promise in their potential role in the maintenance of allograft health and possibly as a tool for the identification of tolerance. More work needs to be done to search for immunosuppressive agents with fewer adverse effects,

optimal treatment for AMR and PCR, and means to prevent recurrence of metabolic and immune-mediated diseases in the graft. Finally, further studies on the use of HBsAg-positive allografts and the economics of machine perfusion can further expand access to the donor pool and translate to more lives saved.

FUNDING INFORMATION

Financial support and sponsorship: American Association for the Study of Liver Diseases.

ACKNOWLEDGMENTS

The authors thank Jenny Castano, Krisie Gardner, Jaime Thomas, Kearin Reid, and Caitlin Bakker for their contributions to the development of this Guideline and the AASLD Practice Guidelines Committee (PGC) for their support and input, including valuable peer review led by Lindsay Y. King. Members of the AASLD PGC include Elizabeth Verna (Chair), Cynthia Levy (Vice Chair), Tamar Taddei (Board Liaison), Therese Bittermann, Po-Hung (Victor) Chen, Kathleen E. Corey, Thomas Cotter, Archita Parikh Desai, Albert Do, Juan F. Gallegos-Orozco, Lindsay Y. King, Christina Cress Lindenmeyer, Anthony J. Michaels, Arpan Mohanty, Andrew Moon, Jennifer C. Price, Elizabeth Rand, Christopher Shubert, Ashwani K. Singal, Puneeta Tandon, and Ethan M. Weinberg.

CONFLICTS OF INTEREST

Vatche G. Agopian consults for Eximium Diagnostics, Merck, and Early Diagnostics, Inc. Paul Y. Kwo consults for Gilead, AbbVie, Aligos, Surrzen, PPD, Novo Nordisk, Hepquant, and Inventiva. Mark W. Russo consults for Lilly. Gerald Scott Winder consults for GlaxoSmithKline. Thomas D. Schiano advises Eurofins/Transplant Genomics, Inc. The remaining authors have no conflicts to report.

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How to cite this article: Te HS, Agopian VG, Demetris AJ, Kwo PY, McGuire BM, Russo MW, et al. AASLD AST Practice Guideline on adult liver transplantation: Diagnosis and management of graft-related complications. *Liver Transpl*. 2026;32:444–490. <https://doi.org/10.1097/LVT.0000000000000715>